

The Jacobian Chain: A Geometric Foundation for Neural Collapse and Trainability

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Abstract

This paper studies a basic question in deep learning: how a trainable network turns raw data into representations that suppress within-class nuisance variation, preserve semantic separation across classes, simplify the final classification problem, and remain stable under optimization. We develop a geometric foundation for that transformation. Our central object is the *Jacobian chain* generated by the network. Its first-order action governs semantic transport and nuisance contraction; its second-order action governs nonlinear distortion and folding; and the quotient-information geometry induced by the model governs meaningful notions of sharpness and stable learning-rate scales.

We formulate a unified framework for both residual networks and Transformers. On the representation side we introduce semantic secants, nuisance fibers, curvature injection, collapse ratios, and late-phase anisotropy defects; on the optimization side we use quotient geometry and Fisher geometry to define symmetry-corrected sharpness and stable optimization scales. The hard core of the paper is a bridge theorem showing that if the Jacobian chain contracts within-class fibers, approximately preserves inter-class semantic secants, and accumulates only a summable curvature floor, then within-class covariance collapses while the between-class semantic skeleton remains nondegenerate. This yields intermediate neural collapse, prototype concentration, and explicit nearest-class-center error bounds.

We then analyze how terminal collapse geometry can arise. Under late-phase isotropization of the surviving semantic covariance, the class means are forced toward regular-simplex / ETF geometry. We prove reduced spectral-selection results for isotropy, connect them back to full-network cross-entropy gradient flow and small-step SGD, and show how primitive regularized head dynamics can make the final classifier–feature self-duality endogenous rather than externally imposed.

For deep learning, the payoff is concrete. The theory explains why useful features become prototype-like without losing class separation, identifies measurable quantities that predict trainability, benign collapse, and instability, and suggests architectural and training interventions for steering networks toward stable propagation and simpler terminal representations. In this way, the paper turns representation formation, collapse, and optimization into parts of one mathematical picture.

1 Introduction

Deep learning works when a network does more than fit labels. It must turn raw data into a representation that is cleaner within each class, better separated across classes, easier to classify, and still trainable under gradient-based optimization. In modern classification systems this often appears as a characteristic late-phase simplification of the learned representation: within-class variability shrinks, class prototypes sharpen, the effective decision rule becomes simpler, and in favorable balanced regimes the geometry approaches neural collapse / regular-simplex / ETF structure. At the same time, the same network must continue to propagate information and gradients stably, control nonlinear distortion, and tolerate finite learning rates. Existing theory usually studies these phenomena with different mathematical languages. The main goal of this paper is to put them into one framework.

Our proposal is that the common object is the *Jacobian chain* generated by the network. A deep model should be viewed not only as a parameterized function, but as a composition of local maps whose first-order action determines semantic transport and nuisance contraction, whose second-order action determines distortion and folding, and whose induced quotient-information geometry determines meaningful sharpness and stable optimization scales. This leads to the organizing principle

$$\begin{aligned} \text{deep network} &\implies \text{Jacobian chain} \\ &\implies \text{geometry of transport, compression, collapse, and optimization.} \end{aligned} \tag{1}$$

The concrete deep-learning question studied here is therefore the following:

How does a trainable deep network transform raw data into classwise denoised, semantically separated, and easy-to-classify features without losing optimization stability?

We answer this question geometrically, using Jacobians, higher jets, semantic secants, nuisance fibers, quotient / Fisher geometry, and late-phase reduced semantic dynamics. The point is not to attach geometric terminology to deep learning after the fact. The point is to identify the geometric variables that explain representation simplification, benign collapse, and late-phase optimization behavior.

Two architectural families motivate the framework. Residual networks admit a continuous-depth and mean-field interpretation [2, 9, 13]. Transformers admit a context-dependent interpretation in which the local transport rule depends on an empirical token measure [7, 21]. Despite their surface differences, both classes must solve the same mathematical problem: build a Jacobian chain that is expressive enough to shape representation, stable enough to propagate signal, and regular enough to remain trainable.

The thesis of the paper can be compressed into one sentence.

The central problem of deep learning is the design of a Jacobian chain that preserves semantic directions, contracts nuisance directions, controls nonlinear distortion, and remains trainable under finite-step optimization.

(2)

What this paper is about. The paper focuses on three linked deep-learning phenomena:

- Q1.** why successful deep representations suppress within-class variability without destroying class-wise semantic separation;
- Q2.** why late training often drives the representation toward a prototype regime and, in balanced settings, further toward neural collapse / simplex / ETF geometry;
- Q3.** which measurable geometric quantities predict trainability, benign collapse, optimization stability, and failure modes.

Main contributions. The manuscript now contains a clear mathematical hard core.

- C1. A unified Jacobian-chain setup for ResNets and Transformers.** We formulate one residualized state update that covers both standard residual flows and context-dependent Transformer flows.
- C2. A bridge theorem from local geometry to intermediate neural collapse.** We prove that semantic-secant preservation together with within-class fiber contraction and a summable curvature floor implies decay of a layerwise collapse ratio, thereby turning local Jacobian geometry into a global statement about learned class statistics.
- C3. Direct deep-learning corollaries.** From the bridge theorem we derive class-conditional prototype concentration, an explicit vanishing-error guarantee for nearest-class-center classification, and—under semantic isotropy—a terminal-rigidity theorem forcing regular-simplex / ETF geometry.
- C4. A late-phase route to terminal geometry.** We show that balanced reduced semantic dynamics select isotropy, that late-phase full-network cross-entropy gradient flow and small-step SGD inherit the same selector up to vanishing error, and that primitive regularized head dynamics can make the final classifier–feature self-duality endogenous.
- C5. A theorem-guided geometric program for deep learning.** Beyond the hard core, we retain supporting propositions, architecture-level conjectures, and a measurement/intervention protocol connecting the core theory to Transformer holonomy, attention spectral gaps, topology change, Hamiltonian transport, and quotient-information optimization geometry.

Why this matters for deep learning. If the program is correct, then several phenomena that are often treated separately—signal propagation, benign feature collapse, prototype formation, terminal neural collapse, and learning-rate geometry—can be read as different consequences of one underlying mechanism. Just as importantly, the theory produces quantities that can be measured or controlled in practice, including semantic secant distortion, nuisance-fiber contraction, curvature floor, anisotropy defect, prototype margin, and quotient / Fisher sharpness. The intended outcome is therefore not only conceptual unification, but also diagnostics and design principles.

Positioning. Some of the Transformer-specific geometric results motivating the broader program are recent preprints rather than settled textbook facts. We therefore use them as motivation and as experimental targets, not as axioms. By contrast, the bridge theorem and its immediate corollaries rest on standard tools from differential geometry, moment transport, matrix analysis, and local stability theory.

Paper organization. Section 2 situates the paper relative to prior work. Section 3 summarizes the conceptual architecture and isolates the hard core. Section 4 introduces the unified Jacobian-chain setup. Section 5 defines the main geometric objects. Section 6 develops the hard theorem package of the manuscript: a bridge theorem from Jacobian-chain geometry to classwise collapse, a rigidity upgrade under semantic isotropization, a late-phase reduction theorem connecting full-network cross-entropy dynamics to reduced semantic selection, and a primitive head-dynamics theorem that makes terminal self-duality endogenous. Section 7 records the broader supporting propositions and architecture-level conjectures. Section 8 supplies proof sketches for the supporting propositions. Sections 9–11 describe the theorem-guided experimental program, observables, and validation logic. Sections 12–14 discuss figures, limitations, open problems, and conclusions.

2 Related Work

This paper sits at the intersection of four literatures that are often studied separately.

Deep residual dynamics and signal propagation. Residual networks [9] and their continuous-depth formulation as Neural ODEs [2] provide the basic viewpoint that depth can be analyzed as the composition of local flows rather than as a purely combinatorial stack of layers. Mean-field analyses of deep residual architectures [13] and signal-propagation theories based on edge-of-chaos and dynamical-isometry ideas [18, 20] show that trainability is tightly tied to the singular-value geometry of the input–output Jacobian. These works motivate our insistence that representation quality and trainability must be read through the Jacobian chain.

Optimization geometry, information geometry, and parameter symmetries. Natural gradient formalizes optimization on a statistical manifold endowed with the Fisher–Rao metric [14]. Path-SGD emphasizes that positively homogeneous networks carry nontrivial rescaling symmetries and should therefore be optimized using geometry that respects those symmetries [16]. Edge-of-stability analyses show that practical deep-learning optimization often runs near a sharpness-controlled threshold [1], while Fisher-SAM and recent symmetry-aware sharpness work for Transformers argue that meaningful sharpness should be defined through information geometry and quotient geometry rather than raw Euclidean Hessians [5, 8, 11]. These lines motivate our use of quotient-information geometry on the optimization side of the theory.

Neural collapse and late-phase representation simplification. Neural collapse was identified empirically by Pappas et al. [17] and has since been formalized from several complementary perspectives. The feature-model landscape analysis of Zhou et al. [22] and the deep-linear analysis of Dang et al. [23] show that simplex / ETF terminal geometry is not merely anecdotal. The perturbative late-phase analysis of Tirer et al. [24] and the intermediate-neural-collapse study of Rangamani et al. [19] show that collapse is a dynamical process, not just a final-layer curiosity. Our bridge theorem is designed to explain this process geometrically: it derives prototype formation and intermediate collapse from local Jacobian-chain conditions.

Transformers, context dependence, and spectral effects. Transformers were introduced in Vaswani et al. [21]. The mathematical framework of Geshkovski et al. [7] models Transformers as interacting particle systems and empirical-measure flows, which is our main formal reason

for treating context as part of the state. More recent work studies trainability in deep random Transformers through geometric dynamics and Lyapunov exponents [4], while spectral analyses of softmax attention identify spectral-gap mechanisms behind rank collapse and signal-propagation pathologies [15]. These results motivate the noncommutative and measure-dependent side of our broader Jacobian-chain program.

Topology and structure-preserving network geometry. Persistent-homology regularization shows that topological quantities can enter learning objectives directly rather than serving only as post hoc diagnostics [3]. On the structure-preserving side, symplectic neural networks and Hamiltonian deep networks show that geometric structure can be used to guarantee long-range stability and suppress gradient decay by design [6, 10]. We use these lines mainly to motivate the extended part of the program: topological complexity, holonomy, and symplectic defect are treated as observables and conjectural explanatory variables around the hard core.

Relation to the present paper. The present paper is not a new result inside any one of these literatures taken in isolation. Its contribution is to connect them through a single mathematical object—the Jacobian chain—and to show that representation simplification, intermediate collapse, terminal simplex rigidity, and optimization geometry can be studied as different consequences of its first-order action, second-order action, noncommutativity, and quotient structure.

3 Overview of the Program and Contributions

From a deep-learning point of view, the paper studies one transformation: how a trainable network turns raw data into features that are easier to classify, less variable within each class, more stable under optimization, and eventually simple enough to enter a prototype regime. The claim of the paper is that this transformation is governed by the Jacobian chain and the geometries induced by its first- and second-order behavior.

The representation-side logic is

$$\begin{aligned} \text{raw data} &\implies \text{deep representation} \implies \text{Jacobian chain} \\ &\implies \left\{ \begin{array}{ll} \text{restricted singular values} & \rightarrow \text{signal propagation and compression,} \\ \text{second-order injection} & \rightarrow \text{folding, distortion, and instability,} \\ \text{commutators} & \rightarrow \text{holonomy and context dependence,} \\ \text{rank loss / noninjectivity} & \rightarrow \text{topological change.} \end{array} \right. \end{aligned} \quad (3)$$

The optimization-side logic is

$$\begin{aligned} \text{parameter symmetries} &\implies \Theta/G, \\ \text{probabilistic outputs} &\implies \mathcal{M}_{\text{stat}}, \\ \text{optimization} &\implies \text{quotient-information geometry.} \end{aligned} \quad (4)$$

Together these two chains explain why trainability, collapse, and late-phase geometry should not be studied as unrelated phenomena.

The hard core of the paper. The manuscript is organized around one central implication:

$$\begin{aligned} \text{local Jacobian geometry} &\implies \text{global class statistics} \implies \text{intermediate collapse} \\ &\implies \text{late semantic anisotropy dissipation} \implies \text{semantic isotropy} \\ &\implies \text{terminal simplex rigidity and self-duality.} \end{aligned}$$

This is the part of the paper intended to carry the main mathematical weight. The surrounding propositions, conjectures, and experiments are an extended program built around this core rather than a flat list of independent ideas.

Core theorem package. The hard core has four layers.

- (H1) **Bridge theorem.** A selective semantic-isometry / fiber-contraction theorem shows that if the Jacobian chain contracts within-class fibers, approximately preserves inter-class semantic secants, and accumulates only a summable curvature floor, then the collapse ratio decays and the representation enters an intermediate-neural-collapse regime.
- (H2) **Deep-learning corollaries.** The bridge theorem yields class-conditional prototype concentration and an explicit nearest-class-center error bound, so the geometric statement immediately translates into a prediction about feature simplification and classification.
- (H3) **Rigidity and selection.** Under semantic isotropy, the surviving class means are forced toward regular-simplex / ETF geometry. Balanced reduced semantic dynamics then select isotropy, and late-phase full-network gradient flow together with small-step SGD inherit this selector up to vanishing error.
- (H4) **Endogenous terminal alignment.** Primitive regularized head dynamics can drive the final classifier into the same semantic span and onto the same self-dual ray, making the terminal classifier–feature alignment part of the dynamics rather than an external assumption.

Why this is useful for deep learning. The theory is designed to be useful in three ways.

- (U1) It gives a mathematical explanation for why good deep features become prototype-like without losing class separation.
- (U2) It identifies measurable quantities—semantic secant distortion, nuisance-fiber contraction, curvature floor, anisotropy defect, prototype margin, quotient sharpness, Fisher sharpness—that can serve as diagnostics for trainability, benign collapse, and instability.
- (U3) It suggests interventions: architecture, normalization, head dynamics, and training schedules can in principle be designed to shape the relevant geometric quantities rather than tuned only by trial and error.

What remains broader than the hard core. Around the theorem package, the paper keeps a broader geometric program involving Transformer holonomy, attention spectral gaps, topology change, Hamiltonian transport, and quotient-information optimization geometry. These parts are important, but they are not meant to compete with the central collapse/rigidity/selection chain. They enlarge the scope of the program once the hard core is in place.

Table 1: Core symbols used in the hard theorem package.

Symbol	Meaning
W_l	Within-class covariance at depth l
B_l	Between-class covariance / semantic skeleton at depth l
ρ_l	Fiber-contraction coefficient along class manifolds
δ_l, ε_l	Local and effective semantic-secant distortion coefficients
η_l	One-step curvature floor / second-order remainder scale
M_t	Centered prototype matrix in late training
$\hat{G}_t$	Normalized semantic Gram matrix associated with M_t
H_t	Linear classifier / head at late-training time t
δ_t^{iso}	Semantic isotropy defect
$\mathcal{A}_t^{\text{head}}$	Classifier-prototype self-dual alignment defect

Table 2: The hard-core theorem packages and what they establish.

Package	Mathematical output
Bridge package	Local Jacobian geometry $\Rightarrow$ intermediate neural collapse
Prototype package	Collapse $\Rightarrow$ prototype concentration and simplified classification
Rigidity package	Semantic isotropy $\Rightarrow$ simplex / ETF terminal rigidity
Selection package	Balanced late-phase dynamics $\Rightarrow$ isotropy selection
Reduction package	Full-network CE flow / SGD $\Rightarrow$ reduced semantic dynamics
Head package	Primitive regularized head flow $\Rightarrow$ endogenous self-dual alignment

Reader’s guide: notation and theorem packages

Two reading conventions are used throughout the manuscript.

- (G1) Symbols indexed by depth l describe *representation geometry* inside the network. In particular, W_l always denotes the within-class covariance.
- (G2) Symbols indexed by continuous time t describe *late-phase optimization dynamics*. In particular, H_t always denotes the linear classifier / head, while M_t denotes the centered prototype matrix and $\hat{G}_t$ the normalized semantic Gram matrix.

This convention is important because the hard-core theorem chain mixes representation statistics, reduced semantic dynamics, and head optimization.

4 Unified Jacobian-Chain Setup

We begin from a single residualized update rule that can represent both ResNet-like and Transformer-like architectures:

$$s_{l+1} = \Phi_l(s_l, \nu_l; \theta_l) = s_l + h_l F_l(s_l, \nu_l; \theta_l), \quad l = 0, \dots, L-1. \quad (5)$$

Here s_l denotes the network state at depth l , θ_l denotes the trainable parameters of layer l , and ν_l encodes the context variable relevant to the architecture.

For a ResNet, one may take

$$s_l = z_l \in \mathbb{R}^{d_l}, \quad \nu_l \equiv \emptyset,$$

so that the update reduces to an ordinary residual flow on a finite-dimensional state space.

For a Transformer with n tokens, it is more natural to write

$$s_l = X_l = (x_l^1, \dots, x_l^n) \in (\mathbb{R}^d)^n, \quad \nu_l = \mu_l := \frac{1}{n} \sum_{i=1}^n \delta_{x_l^i},$$

where μ_l is the empirical token measure. In the mean-field viewpoint, the architecture is thus naturally interpreted as a map on a space of probability measures, implemented by an interacting particle system [7].

This observation motivates a shift of viewpoint. The network should not be studied merely as a composition of nonlinear maps, but as a chain of differential objects,

$$\left\{ j^1 \Phi_l, j^2 \Phi_l, [j^1 \Phi_i, j^1 \Phi_j], \text{rank}(j^1 \Phi_l) \right\}_{l=0}^{L-1}, \quad (6)$$

where

- the first jet controls lengths, angles, volumes, conditioning, and gradient propagation;
- the second jet controls bending, curvature injection, and folding;
- noncommutativity controls connection curvature, holonomy, and path dependence;
- rank loss and noninjectivity control singularity structure and topological change.

In the sequel we will often suppress the dependence of Φ_l on (ν_l, θ_l) in order to focus on the geometric dependence on the state variable. For an input point $x \in M_0$, we write

$$z_l(x) := \Phi_{l-1} \circ \dots \circ \Phi_0(x) \in M_l$$

for its representation at depth l .

5 Definitions

This section introduces the geometric objects that will serve as the native language of the program.

Definition 5.1 (Data manifold and layerwise representation manifolds). *Assume the input data are concentrated on an m -dimensional manifold $M_0 \subset \mathbb{R}^{d_0}$. Define the layerwise representation manifold by*

$$M_l = \Phi_{l-1} \circ \dots \circ \Phi_0(M_0). \quad (7)$$

If the data are class-labeled, we denote by $M_l^c \subset M_l$ the class- c representation manifold at depth l .

Definition 5.2 (Task bundle and nuisance bundle). *For each $x \in M_0$, choose a local splitting*

$$T_x M_0 = T_x^{\text{task}} \oplus N_x^{\text{nuis}}, \quad (8)$$

where T_x^{task} is the task-relevant tangent subspace and N_x^{nuis} is the nuisance subspace along which label-conditional statistics are approximately invariant. If a global direct-sum decomposition does not exist, the splitting is understood locally. At layer l , we write the propagated splitting as

$$T_{z_l} M_l = T_l^{\text{task}} \oplus N_l^{\text{nuis}}. \quad (9)$$

Definition 5.3 (Jacobian chain). *Let*

$$J_l := D_{s_l} \Phi_l(s_l, \nu_l; \theta_l) \quad (10)$$

denote the state Jacobian of layer l . For integers $b > a$, define the partial Jacobian chain

$$\mathcal{J}_{b:a} := J_{b-1} J_{b-2} \cdots J_a. \quad (11)$$

In particular, the Jacobian from input to layer l is

$$\mathcal{J}_{l:0} = J_{l-1} \cdots J_0, \quad \mathcal{J}_{a:a} := I. \quad (12)$$

For measure-dependent architectures, J_l is understood either as the full block Jacobian on token space $(\mathbb{R}^d)^n$ or, in the mean-field limit, as the Fréchet derivative of the induced flow map on $\mathcal{P}_2(\mathbb{R}^d)$.

Definition 5.4 (Selective isometry indices). *For the task and nuisance subspaces, define*

$$\alpha_l(x) := \inf_{\substack{v \in T_x^{\text{task}} \\ \|v\|=1}} \|\mathcal{J}_{l:0}(x)v\|, \quad \beta_l(x) := \sup_{\substack{w \in N_x^{\text{nuis}} \\ \|w\|=1}} \|\mathcal{J}_{l:0}(x)w\|. \quad (13)$$

Define the task-restricted condition number by

$$\kappa_l^{\text{task}}(x) := \frac{\sigma_{\max}(\mathcal{J}_{l:0}|_{T_x^{\text{task}}})}{\sigma_{\min}(\mathcal{J}_{l:0}|_{T_x^{\text{task}}})}. \quad (14)$$

A network achieves selective isometry if, over a substantial depth range,

$$\alpha_l \approx 1, \quad \beta_l \ll 1. \quad (15)$$

Remark 5.5. *Definition 5.4 deliberately differs from full dynamical isometry. The objective is not to preserve all directions equally, but to preserve task-relevant directions while suppressing nuisance directions. This is a more task-aligned geometric target than full-space isometry [18].*

Definition 5.6 (Local strain and rotation). *For a residual block*

$$\Phi_l(z) = z + h_l f_l(z), \quad A_l := Df_l(z_l), \quad (16)$$

so that $J_l = I + h_l A_l$, define the symmetric and skew-symmetric parts

$$S_l := \frac{A_l + A_l^\top}{2}, \quad \Omega_l := \frac{A_l - A_l^\top}{2}. \quad (17)$$

We interpret S_l as the local strain tensor and Ω_l as the local rotation or twisting tensor.

Definition 5.7 (Curvature injection). *Assume Φ_l is C^2 . The task-restricted second-order nonlinear strength of layer l is*

$$\chi_l(x) := \left\| D^2 \Phi_l(z_l) \Big|_{T_l^{\text{task}} \times T_l^{\text{task}}} \right\|_{\text{op}}. \quad (18)$$

Equivalently, one may consider the directional version

$$\chi_l(x; u, v) := \left\| D^2 \Phi_l(z_l)[u, v] \right\|. \quad (19)$$

This quantity measures not an abstract ambient curvature, but the amount of new bending injected by layer l itself.

Definition 5.8 (Accumulated curvature injection). *Define the cumulative curvature injection from input to layer L by*

$$\mathfrak{C}_L(x) := \sum_{l=0}^{L-1} \|\mathcal{J}_{L:l+1}(x_l)\|_{\text{op}} \chi_l(x_l) \|\mathcal{J}_{l:0}(x)\|_{\text{op}}^2. \quad (20)$$

This quantity measures how much second-order bending is injected at layer l and then amplified by the surrounding Jacobian chain.

Definition 5.9 (Holonomy and noncommutativity energy). *For residual generators A_l , define the discrete holonomy energy*

$$\mathfrak{H}_L := \sum_{0 \leq i < j \leq L-1} \|[A_i, A_j]\|_F^2, \quad [A_i, A_j] = A_i A_j - A_j A_i. \quad (21)$$

A task-restricted version is

$$\mathfrak{H}_L^{\text{task}} := \sum_{i < j} \|P_i[A_i, A_j]P_i\|_F^2, \quad (22)$$

where P_i denotes the orthogonal projection onto T_i^{task} .

For context-dependent architectures such as Transformers, define a context holonomy by

$$\mathfrak{H}_l^{\text{ctx}}(x) := \left\| P_l(J_l^{(c_1)} J_l^{(c_2)} - J_l^{(c_2)} J_l^{(c_1)}) P_{l-1} \right\|_F^2, \quad (23)$$

where c_1 and c_2 denote two context perturbations. This quantity is intended to capture whether the same representation direction is transported differently under different context paths.

Definition 5.10 (Parameter quotient space and horizontal sharpness). *Let Θ denote the parameter space and let G denote the symmetry group preserving the realized network function. Define the quotient space*

$$\tilde{\Theta} := \Theta/G. \quad (24)$$

At $\theta \in \Theta$, decompose the tangent space as

$$T_\theta \Theta = \mathcal{V}_\theta \oplus \mathcal{H}_\theta, \quad (25)$$

where $\mathcal{V}_\theta$ is the vertical space tangent to the symmetry orbit and $\mathcal{H}_\theta$ is a chosen horizontal complement. If H_θ is the Hessian and F_θ the Fisher information matrix, define the horizontal sharpness

$$\lambda_{\max}^{\text{hor}}(H_\theta) := \sup_{\substack{\xi \in \mathcal{H}_\theta \\ \|\xi\|=1}} \langle \xi, H_\theta \xi \rangle. \quad (26)$$

Remark 5.11. *Path-SGD already suggests that rescaling-invariant geometry is more natural than raw Euclidean geometry for positively homogeneous networks [16]. Recent symmetry-aware sharpness work for Transformers sharpens this point by showing that sharpness should be measured on an appropriate quotient manifold rather than on the raw parameter space [5].*

Definition 5.12 (Statistical manifold and Fisher geometry). *Let*

$$\mathcal{M}_{\text{stat}} = \{p_\theta(y | x) : \theta \in \Theta\} \quad (27)$$

denote the statistical manifold realized by the model. Its local geometry is endowed with the Fisher–Rao metric. An optimization step of the form

$$\Delta\theta = -F_\theta^{-1} \nabla_\theta L \quad (28)$$

is then the natural-gradient direction on $\mathcal{M}_{\text{stat}}$ [14].

Definition 5.13 (Topological complexity of intermediate representations). *For each class c and depth l , define the persistence diagram*

$$\text{Dgm}_k(M_l^c) \quad (29)$$

and the corresponding Betti curve

$$\beta_k^\varepsilon(M_l^c). \quad (30)$$

These quantities quantify layerwise topological complexity such as connected components, loops, and higher-dimensional holes. Persistent-homology regularization has already been used directly as a learning principle, showing that topology here is not merely interpretive language [3].

Definition 5.14 (Symplectic defect). *Suppose a sub-block acts on paired variables $z = (q, p)$ carrying a symplectic form represented by a matrix Ω . Define the symplectic defect of layer l by*

$$\delta_l^{\text{symp}} := \left\| J_l^\top \Omega J_l - \Omega \right\|_F. \quad (31)$$

When $\delta_l^{\text{symp}} \approx 0$, layer l is approximately symplectic. This is motivated by Hamiltonian deep networks and SympNets, where symplectic structure directly controls gradient propagation and long-range transport [6, 10].

6 The Hard Theorem Package: From Jacobian Geometry to Neural Collapse

The previous sections introduced the Jacobian-chain framework and the geometric observables that will be used throughout the paper. This section is the mathematical center of the manuscript. Its purpose is to answer, in theorem form, the main deep-learning question announced in the introduction: why do successful deep classifiers produce classwise denoised, semantically separated, and progressively simplified representations, and why can that process culminate in neural-collapse geometry rather than in indiscriminate rank destruction?

The answer developed here has four layers. First, a bridge theorem shows that local Jacobian geometry can force a representation-level phenomenon: contraction along within-class fibers, near-isometry on inter-class semantic secants, and summable curvature error imply intermediate collapse. Second, representation-side corollaries show what this buys in deep-learning terms: prototype concentration and simplified classification. Third, a rigidity theorem shows that once the surviving semantic covariance becomes isotropic, the remaining class geometry is no longer flexible and must approach regular-simplex / ETF form. Fourth, a sequence of late-phase selection theorems connects that isotropy and the eventual classifier-feature alignment back to reduced and then full-network training dynamics.

A fully rigorous final version would require explicit regularity, concentration, and measurability hypotheses; these are deferred to the appendix. What matters here is the logical shape of the theorem package. The statements below are not a flat collection of estimates. For orientation, Table 2 should be read as the dependency ledger of this section: the bridge package produces collapse, the rigidity package upgrades collapse to simplex structure, the selection package explains why isotropy is chosen, the reduction package connects that selector back to full training, and the head package internalizes the final classifier alignment. They form a chain

$$\begin{aligned} \text{local Jacobian geometry} &\implies \text{classwise collapse} \implies \text{prototype concentration} \\ &\implies \text{semantic isotropy} \implies \text{simplex rigidity and self-duality.} \end{aligned}$$

That chain is the hard core around which the rest of the paper is organized.

Remark 6.1 (Standing notation for the hard core). *Depth-indexed quantities l refer to representation formation across layers. Continuous-time quantities t refer to late-phase optimization after a prototype regime has emerged. We reserve W_l for within-class covariance and H_t for classifier / head parameters. This separation prevents notational collisions between representation statistics and optimization variables in the later reduction and head-dynamics theorems.*

6.1 Classwise geometry: nuisance fibers and semantic secants

We begin by replacing the earlier abstract task/nuisance splitting with a classification-oriented geometric decomposition. The correct objects are class fibers and semantic secants.

Definition 6.2 (Classwise representation statistics). *Let the input distribution be supported on a disjoint union of class-conditional supports*

$$M_0 = \bigsqcup_{c=1}^K M_0^c \subset \mathbb{R}^{d_0},$$

and let

$$f_l = \Phi_{l-1} \circ \cdots \circ \Phi_0, \quad Z_l = f_l(X), \quad M_l^c = f_l(M_0^c).$$

For each class c , define the class mean and class covariance by

$$\mu_l^c := \mathbb{E}[Z_l \mid Y = c], \quad \Sigma_l^c := \text{Cov}(Z_l \mid Y = c). \quad (32)$$

Define the global mean, the within-class covariance, and the between-class covariance by

$$\bar{\mu}_l := \frac{1}{K} \sum_{c=1}^K \mu_l^c, \quad W_l := \frac{1}{K} \sum_{c=1}^K \Sigma_l^c, \quad B_l := \frac{1}{K} \sum_{c=1}^K (\mu_l^c - \bar{\mu}_l)(\mu_l^c - \bar{\mu}_l)^\top. \quad (33)$$

The matrix W_l measures within-class variability, while B_l measures the semantic skeleton carried by class means.

Definition 6.3 (Fiber geometry and semantic secants). *For a point $z \in M_l^c$, the class fiber tangent space is*

$$N_z^l := T_z M_l^c. \quad (34)$$

The set of semantic secants at layer l is

$$\mathcal{S}_l := \{\mu_l^c - \mu_l^{c'} : c \neq c'\}, \quad E_l^{\text{sem}} := \text{span}(\mathcal{S}_l). \quad (35)$$

For $z \in M_l^c$, define the local semantic secant cone

$$\mathcal{C}_l^{\text{sem}}(z) := \text{cone}\{\mu_l^{c'} - z : c' \neq c\}. \quad (36)$$

Thus the geometric mechanism relevant for classification is twofold: contraction along class fibers N_z^l , and stability of semantic secants in E_l^{sem} .

Definition 6.4 (Layerwise contraction, semantic distortion, and curvature floor). *For each layer l , define the Jacobian and curvature magnitudes*

$$J_l(z) := D\Phi_l(z), \quad \chi_l := \sup_{z \in \bigcup_{c=1}^K M_l^c} \|D^2\Phi_l(z)\|_{\text{op}}, \quad J_l := \sup_{z \in \bigcup_{c=1}^K M_l^c} \|J_l(z)\|_{\text{op}}. \quad (37)$$

Define the fourth-moment class radius by

$$r_l := \max_{1 \leq c \leq K} \left(\mathbb{E}[\|Z_l - \mu_l^c\|^4 \mid Y = c] \right)^{1/4}. \quad (38)$$

The fiber contraction coefficient is

$$\rho_l := \sup_{1 \leq c \leq K} \sup_{z \in M_l^c} \sup_{\substack{u \in N_z^l \\ \|u\|=1}} \|J_l(z)u\|. \quad (39)$$

The local semantic distortion coefficient is the smallest $\delta_l \geq 0$ such that

$$(1 - \delta_l) \|v\| \leq \|J_l(z)v\| \leq (1 + \delta_l) \|v\| \quad \text{for all } z \in M_l^c, v \in \mathcal{C}_l^{\text{sem}}(z), c = 1, \dots, K. \quad (40)$$

Finally, define the curvature floor

$$\eta_l := C_\eta (J_l \chi_l r_l^3 + \chi_l^2 r_l^4), \quad (41)$$

where $C_\eta > 0$ is a universal constant.

Remark 6.5 (Why this geometry is sharper than a tangent-bundle-only picture). *For continuous nuisance variables, tangent-bundle language is natural. For classification, however, discriminative information is not primarily stored in tangent directions inside a class manifold; it is stored in inter-class secants. A deep classification network succeeds when it contracts the class fibers while preserving the semantic secants. This is the geometric content of the theorem below.*

6.2 Bridge theorem: selective compression yields intermediate collapse

The next theorem is the intended hard core of the framework. It is a bridge theorem in the strong sense: it derives a nontrivial representation-level phenomenon from differential-geometric assumptions on the Jacobian chain.

Theorem 6.6 (Selective semantic isometry plus fiber contraction implies collapse). *Assume that each class support M_0^c is a compact C^2 submanifold of $\mathbb{R}^{d_0}$, that each layer map Φ_l is C^2 on a neighborhood of $\bigcup_c M_l^c$, and that the push-forward class supports*

$$M_l^c = f_l(M_0^c) \quad (c = 1, \dots, K)$$

remain embedded C^2 submanifolds.

Suppose that for every layer $l \geq 0$ the following hold:

(A1) Uniform fiber contraction. *One has*

$$\rho_l < 1. \quad (42)$$

(A2) Local semantic secant near-isometry. The coefficient δ_l in (40) satisfies

$$0 \leq \delta_l < 1. \quad (43)$$

(A3) Summable curvature floor. The curvature error term is summable:

$$\sum_{l=0}^{\infty} \eta_l < \infty. \quad (44)$$

(A4) Nondegenerate initial semantic skeleton. The initial between-class covariance is nontrivial on its semantic span:

$$\lambda_{\min}^+(B_0) > 0, \quad (45)$$

where $\lambda_{\min}^+(B_0)$ denotes the smallest nonzero eigenvalue of B_0 on E_0^{sem} .

Then there exists a universal constant $C_\varepsilon > 0$ such that, if one defines

$$\varepsilon_l := \delta_l + C_\varepsilon \chi_l r_l, \quad (46)$$

and assumes $\varepsilon_l < 1$ for all l , the following one-step recursions hold:

$$\text{Tr}(W_{l+1}) \leq \rho_l^2 \text{Tr}(W_l) + \eta_l, \quad (47)$$

and

$$\lambda_{\min}^+(B_{l+1}) \geq (1 - \varepsilon_l)^2 \lambda_{\min}^+(B_l) - \eta_l. \quad (48)$$

Consequently, if in addition

$$\sup_{l \geq 0} \rho_l \leq \bar{\rho} < 1, \quad \sum_{l=0}^{\infty} \varepsilon_l < \infty, \quad \sum_{l=0}^{\infty} \eta_l < \left(\prod_{l=0}^{\infty} (1 - \varepsilon_l)^2 \right) \lambda_{\min}^+(B_0), \quad (49)$$

then

$$\text{Tr}(W_l) \rightarrow 0, \quad (50)$$

while

$$\inf_{l \geq 0} \lambda_{\min}^+(B_l) > 0. \quad (51)$$

In particular, the collapse ratio

$$\text{CR}_l := \frac{\text{Tr}(W_l)}{\lambda_{\min}^+(B_l)} \quad (52)$$

satisfies

$$\text{CR}_l \rightarrow 0. \quad (53)$$

Remark 6.7 (Interpretation). Theorem 6.6 states that collapse is not merely a terminal optimizer artifact. It is the statistical shadow of a Jacobian chain that contracts within-class fibers while preserving inter-class semantic secants, with second-order curvature effects small enough not to destroy the semantic skeleton. In compact form:

$$\begin{aligned} & \text{semantic secant preservation} + \text{fiber contraction} \\ & + \text{summable curvature floor} \implies \text{intermediate collapse.} \end{aligned}$$

This is the sense in which the theorem serves as a genuine geometric mechanism theorem rather than a purely kinematic estimate.

6.3 First collapse corollaries

The bridge theorem answers the mechanism question. The next corollaries restate its conclusion in the language most natural for representation learning: within-class scatter becomes negligible relative to the surviving semantic skeleton, and the learned representation begins to behave like a class-prototype geometry rather than a diffuse feature cloud.

Corollary 6.8 (Intermediate neural collapse). *Under the hypotheses of Theorem 6.6, one has*

$$\frac{\text{Tr}(W_l)}{\text{Tr}(B_l)} \rightarrow 0, \quad \frac{\text{Tr}(W_l)}{\text{Tr}(W_l) + \text{Tr}(B_l)} \rightarrow 0. \quad (54)$$

Hence within-class covariance becomes negligible relative to between-class covariance along depth. In particular, deeper layers move toward the intermediate neural-collapse regime observed empirically in modern networks [19].

Proof sketch. Since $\text{Tr}(B_l) \geq \lambda_{\min}^+(B_l)$ on the nontrivial semantic span and Theorem 6.6 gives

$$\frac{\text{Tr}(W_l)}{\lambda_{\min}^+(B_l)} \rightarrow 0,$$

we immediately obtain

$$\frac{\text{Tr}(W_l)}{\text{Tr}(B_l)} \leq \frac{\text{Tr}(W_l)}{\lambda_{\min}^+(B_l)} \rightarrow 0.$$

The second ratio then follows by dividing numerator and denominator by $\text{Tr}(B_l)$. $\square$

Corollary 6.9 (Emergence of a nearest-class-center regime). *Assume the hypotheses of Theorem 6.6. Define the class support radius*

$$R_l := \max_{1 \leq c \leq K} \sup_{z \in M_l^c} \|z - \mu_l^c\|. \quad (55)$$

Suppose in addition that

$$R_{l+1} \leq \rho_l R_l + C_R \chi_l R_l^2 \quad (56)$$

for some universal constant $C_R > 0$, and that $R_l \rightarrow 0$. Then there exists l_0 such that for every $l \geq l_0$, every class c , and every $z \in M_l^c$,

$$\|z - \mu_l^c\| < \|z - \mu_l^{c'}\| \quad \text{for all } c' \neq c. \quad (57)$$

Thus the nearest-class-center rule is eventually correct on the class supports.

Proof sketch. By the local semantic near-isometry and the summability of (ε_l) , all pairwise class-mean distances remain bounded away from zero:

$$\|\mu_l^c - \mu_l^{c'}\| \geq \left(\prod_{t=0}^{l-1} (1 - \varepsilon_t) \right) \|\mu_0^c - \mu_0^{c'}\|.$$

Since $R_l \rightarrow 0$, for sufficiently large l one has

$$2R_l < \min_{c \neq c'} \|\mu_l^c - \mu_l^{c'}\|.$$

The triangle inequality then gives (57). $\square$

6.4 Prototype concentration and simplified classification

The next pair of corollaries spells out what Theorem 6.6 means in deep-learning terms. Depth acts as classwise denoising and prototype formation, and once class centers remain separated the resulting representation admits an explicit error bound for a simple nearest-class-center classifier.

Corollary 6.10 (Class-conditional prototype concentration). *Under the hypotheses of Theorem 6.6 together with the global conditions in (49), each class-conditional covariance collapses individually:*

$$\text{Tr}(\Sigma_l^c) \rightarrow 0 \quad \text{for every } c = 1, \dots, K. \quad (58)$$

Equivalently,

$$W_2^2(\text{Law}(Z_l \mid Y = c), \delta_{\mu_l^c}) = \text{Tr}(\Sigma_l^c) \rightarrow 0. \quad (59)$$

Hence the learned feature distribution of each class converges, in quadratic Wasserstein distance, to a class prototype.

Proof sketch. The classwise moment-transport lemma in the proof architecture yields, for each fixed class c ,

$$\text{Tr}(\Sigma_{l+1}^c) \leq \rho_l^2 \text{Tr}(\Sigma_l^c) + \eta_l.$$

Under (49), discrete Grönwall implies (58). For any probability measure ν and any point mass δ_m ,

$$W_2^2(\nu, \delta_m) = \int \|z - m\|^2 d\nu(z).$$

Taking $\nu = \text{Law}(Z_l \mid Y = c)$ and $m = \mu_l^c$ gives (59). $\square$

Corollary 6.11 (Prototype-classifier error bound). *Define the nearest-class-center classifier on layer- l features by*

$$\hat{Y}_l^{\text{NCC}}(z) := \arg \min_{1 \leq c \leq K} \|z - \mu_l^c\|. \quad (60)$$

Let

$$d_l := \min_{c \neq c'} \|\mu_l^c - \mu_l^{c'}\|. \quad (61)$$

If $\inf_{l \geq 0} d_l \geq d_* > 0$, then for every class c one has the conditional error bound

$$\mathbb{P}(\hat{Y}_l^{\text{NCC}}(Z_l) \neq Y \mid Y = c) \leq \frac{4}{d_*^2} \text{Tr}(\Sigma_l^c). \quad (62)$$

Consequently,

$$\mathbb{P}(\hat{Y}_l^{\text{NCC}}(Z_l) \neq Y) \leq \frac{4}{d_*^2} \max_{1 \leq c \leq K} \text{Tr}(\Sigma_l^c), \quad (63)$$

and in the balanced-class case this simplifies to

$$\mathbb{P}(\hat{Y}_l^{\text{NCC}}(Z_l) \neq Y) \leq \frac{4}{d_*^2} \text{Tr}(W_l). \quad (64)$$

In particular, under Theorem 6.6 and Corollary 6.10,

$$\mathbb{P}(\hat{Y}_l^{\text{NCC}}(Z_l) \neq Y) \rightarrow 0. \quad (65)$$

Proof sketch. Fix a class c . If $\|Z_l - \mu_l^c\| < d_l/2$, then Z_l is closer to its own class center than to any other class center, so nearest-class-center classification is correct. Therefore

$$\mathbb{P}(\hat{Y}_l^{\text{NCC}}(Z_l) \neq Y \mid Y = c) \leq \mathbb{P}\left(\|Z_l - \mu_l^c\| \geq \frac{d_l}{2} \mid Y = c\right).$$

Applying Markov's inequality to $\|Z_l - \mu_l^c\|^2$ gives

$$\mathbb{P}\left(\|Z_l - \mu_l^c\| \geq \frac{d_l}{2} \mid Y = c\right) \leq \frac{4}{d_l^2} \mathbb{E}[\|Z_l - \mu_l^c\|^2 \mid Y = c] = \frac{4}{d_l^2} \text{Tr}(\Sigma_l^c).$$

Using $d_l \geq d_*$ yields (62); the global bounds follow by averaging over classes, and (65) follows from Corollary 6.10. $\square$

Deep-learning reading. Corollary 6.10 says that a successful deep classifier does not merely “extract features”: it progressively denoises each class toward a prototype. Corollary 6.11 then says that this geometric denoising has a concrete predictive consequence: once the semantic skeleton stays separated, a very simple prototype classifier already achieves vanishing error on the learned representation. For deep learning, the theorem therefore identifies three actionable geometric objectives: make ρ_l small, keep semantic distortion and class-center separation under control, and prevent the curvature floor η_l from accumulating.

6.5 Terminal rigidity of the semantic skeleton

The bridge theorem explains why class clouds contract. What it does not yet explain is why the surviving class means should organize themselves into the very specific simplex / ETF geometry seen in terminal neural collapse. The next theorem isolates that additional step: once the residual semantic covariance becomes isotropic on its surviving span, the remaining geometry rigidifies.

The preceding corollaries explain why depth performs classwise denoising and why a simple prototype rule becomes accurate. To connect the hard core theorem more directly to the terminal-neural-collapse literature, however, one wants a stronger rigidity statement: not merely that within-class spread disappears, but that the surviving semantic skeleton becomes geometrically rigid. The natural rigidity target is a regular simplex, equivalently an equal-norm tight frame (ETF), formed by the centered class means.

For each layer l , define the centered class means and their matrix by

$$m_l^c := \mu_l^c - \bar{\mu}_l, \quad \bar{\mu}_l := \frac{1}{K} \sum_{c=1}^K \mu_l^c, \quad M_l := [m_l^1, \dots, m_l^K] \in \mathbb{R}^{d_l \times K}. \quad (66)$$

Then $\sum_{c=1}^K m_l^c = 0$, hence

$$M_l \mathbf{1} = 0, \quad (67)$$

and the between-class covariance may be written as

$$B_l = \frac{1}{K} M_l M_l^\top. \quad (68)$$

The next theorem shows that once collapse has eliminated within-class variation, an additional asymptotic isotropy of the residual semantic covariance forces the centered class means into a regular simplex.

Theorem 6.12 (Terminal simplex rigidity from semantic isotropization). *Assume the hypotheses of Theorem 6.6 together with the global conditions in (49). Suppose in addition that there exists $l_0 \geq 0$ such that for all $l \geq l_0$ the following hold:*

(B1) Full semantic rank.

$$\text{rank}(B_l) = K - 1. \quad (69)$$

(B2) Asymptotic semantic isotropy. *There exist scalars $\beta_l > 0$ such that*

$$\delta_l^{\text{iso}} := \frac{\|B_l - \beta_l P_l^{\text{sem}}\|_{\text{op}}}{\beta_l} \rightarrow 0, \quad (70)$$

where P_l^{sem} denotes the orthogonal projector onto

$$E_l^{\text{sem}} = \text{Im}(B_l).$$

Then

$$\left\| \frac{1}{K\beta_l} M_l^\top M_l - \Pi_K \right\|_{\text{op}} \rightarrow 0, \quad \Pi_K := I_K - \frac{1}{K} \mathbf{1}\mathbf{1}^\top. \quad (71)$$

Consequently, for every class c ,

$$\|m_l^c\|^2 = \beta_l(K - 1) + o(\beta_l), \quad (72)$$

and for every pair $c \neq c'$,

$$\langle m_l^c, m_l^{c'} \rangle = -\beta_l + o(\beta_l). \quad (73)$$

Equivalently,

$$\max_{c \neq c'} \left| \left\langle \frac{m_l^c}{\|m_l^c\|}, \frac{m_l^{c'}}{\|m_l^{c'}\|} \right\rangle + \frac{1}{K-1} \right| \rightarrow 0. \quad (74)$$

Hence the normalized centered class means form an asymptotic regular simplex, or equivalently an ETF-type geometry, on the $(K - 1)$ -dimensional semantic span.

Proof sketch. Let $G_l := M_l^\top M_l$. Since $B_l = \frac{1}{K} M_l M_l^\top$, the nonzero eigenvalues of G_l are exactly K times the nonzero eigenvalues of B_l . Assumption (70) therefore implies that all nonzero eigenvalues of $G_l/(K\beta_l)$ lie in the interval $[1 - \delta_l^{\text{iso}}, 1 + \delta_l^{\text{iso}}]$. On the other hand, (67) implies $\mathbf{1} \in \ker(G_l)$, and the full-rank condition (69) implies that for all sufficiently large l this is the entire kernel. Therefore $G_l/(K\beta_l)$ converges in operator norm to the unique orthogonal projector onto $\mathbf{1}^\perp$, namely Π_K . Reading off diagonal and off-diagonal entries gives (72)–(74). $\square$

Corollary 6.13 (ETF-type terminal neural collapse under classifier alignment). *Assume the hypotheses of Theorem 6.12. Let $w_l^1, \dots, w_l^K \in \mathbb{R}^{d_l}$ denote class-specific linear classifier weights at layer l , and suppose that*

$$\max_{1 \leq c \leq K} \left\| \frac{w_l^c}{\|w_l^c\|} - \frac{m_l^c}{\|m_l^c\|} \right\| \rightarrow 0. \quad (75)$$

Then the normalized classifier weights also form an asymptotic regular simplex, and the classifier-feature pair becomes asymptotically self-dual:

$$\max_{1 \leq c, c' \leq K} \left| \left\langle \frac{w_l^c}{\|w_l^c\|}, \frac{m_l^{c'}}{\|m_l^{c'}\|} \right\rangle - \Gamma_{cc'} \right| \rightarrow 0, \quad (76)$$

where

$$\Gamma := \frac{K}{K-1} I_K - \frac{1}{K-1} \mathbf{1}\mathbf{1}^\top$$

has diagonal entries 1 and off-diagonal entries $-1/(K-1)$. In particular, the feature means and classifier directions approach the same simplex-ETF geometry characteristic of terminal neural collapse.

Proof sketch. By Theorem 6.12, the normalized centered class means form an asymptotic regular simplex. The alignment hypothesis (75) transfers the same pairwise angle asymptotics to the normalized classifier weights. The cross-Gram matrix between normalized weights and normalized feature means therefore converges entrywise to Γ , which yields (76). $\square$

Deep-learning reading. Theorem 6.6 says that depth drives classwise denoising and prototype formation. Theorem 6.12 goes one step further: once the residual semantic covariance becomes asymptotically isotropic on the surviving semantic span, the remaining geometry is no longer flexible. It rigidifies into a regular simplex. Corollary 6.13 then identifies the standard terminal-neural-collapse picture familiar in deep learning: class means and classifier directions approach the same ETF-type object. From the standpoint of deep learning, this turns terminal collapse from a phenomenological observation into a geometric prediction with a clear missing ingredient: not just contraction, but isotropization of the residual semantic covariance.

6.6 Training-side selection of semantic isotropy

The rigidity theorem shows what happens if isotropy is already present. The next question is therefore a training question: why should late optimization select isotropy rather than some other residual semantic shape? The next three subsections answer that question at increasing levels of realism, beginning with a reduced spectral principle and ending with direct full-flow and head-dynamics statements.

Theorem 6.12 turns semantic isotropy into a clean sufficient condition for simplex / ETF rigidity. A natural next question is whether that isotropy can itself be derived, at least in a reduced late-training regime, from symmetry and optimization principles rather than inserted by hand.

After the prototype-concentration regime of Corollary 6.10, the residual semantic geometry is naturally encoded by the centered class-mean Gram matrix

$$G_l := M_l^\top M_l \in \mathbb{S}_+^K, \quad G_l \mathbf{1} = 0, \quad (77)$$

where $M_l = [m_l^1, \dots, m_l^K]$ is defined in (66). In balanced late training, several reduced models effectively treat the class means, or their Gram matrix, as the surviving semantic degrees of freedom [22–24]. The theorem below isolates a sharp selection principle on that reduced space.

For $\tau > 0$, define the compact convex set

$$\mathcal{G}_\tau := \{G \in \mathbb{S}_+^K : G\mathbf{1} = 0, \text{Tr}(G) = \tau\}. \quad (78)$$

For $G \in \mathcal{G}_\tau$, let

$$\lambda^+(G) = (\lambda_1(G), \dots, \lambda_{K-1}(G)) \in \mathbb{R}_+^{K-1} \quad (79)$$

be its nonzero eigenvalues in nonincreasing order, and define the uniform spectrum

$$\nu_\tau := \left(\frac{\tau}{K-1}, \dots, \frac{\tau}{K-1} \right) \in \mathbb{R}_+^{K-1}. \quad (80)$$

Theorem 6.14 (Balanced spectral selection of semantic isotropy). *Fix $\tau > 0$ and let $\Phi : \mathcal{G}_\tau \rightarrow \mathbb{R}$ be a spectral functional of the form*

$$\Phi(G) = \psi(\lambda^+(G)), \quad (81)$$

where ψ is symmetric, convex, and strictly Schur-convex on

$$\Lambda_\tau := \left\{ \lambda \in \mathbb{R}_+^{K-1} : \sum_{i=1}^{K-1} \lambda_i = \tau \right\}. \quad (82)$$

Then:

(C1) *the unique global minimizer of Φ on $\mathcal{G}_\tau$ is*

$$G_\tau^{\text{iso}} := \frac{\tau}{K-1} \Pi_K, \quad \Pi_K := I_K - \frac{1}{K} \mathbf{1}\mathbf{1}^\top; \quad (83)$$

(C2) *every local minimizer of Φ on $\mathcal{G}_\tau$ coincides with G_τ^{iso} .*

Consequently, any gradient-like late-training dynamics on $\mathcal{G}_\tau$ whose Lyapunov functional is Φ and whose limit is a local minimizer must satisfy

$$G_t \rightarrow G_\tau^{\text{iso}}. \quad (84)$$

Proof sketch. The argument has four steps.

Step 1: reduction to a spectral simplex. Every $G \in \mathcal{G}_\tau$ is positive semidefinite, annihilates $\mathbf{1}$, and has trace τ . Hence its nonzero spectrum belongs to the simplex Λ_τ in (82). Conversely, every $\lambda \in \Lambda_\tau$ is realized by some $G \in \mathcal{G}_\tau$, obtained by placing λ on the $(K-1)$ -dimensional subspace $\mathbf{1}^\perp$ and zero on $\text{span}\{\mathbf{1}\}$.

Step 2: majorization. Let $\nu_\tau = (\tau/(K-1), \dots, \tau/(K-1))$. For every $\lambda \in \Lambda_\tau$, the partial sums of the nonincreasing rearrangement satisfy

$$\sum_{i=1}^m \lambda_i \geq \sum_{i=1}^m \frac{\tau}{K-1} \quad (1 \leq m \leq K-1),$$

with equality at $m = K-1$. Thus $\lambda \succ \nu_\tau$ in the Hardy–Littlewood–Pólya order. Since ψ is symmetric, convex, and strictly Schur-convex, one has

$$\psi(\lambda) \geq \psi(\nu_\tau),$$

with equality if and only if $\lambda = \nu_\tau$.

Step 3: identification of the minimizer. Applying Step 2 to $\lambda = \lambda^+(G)$ shows that every $G \in \mathcal{G}_\tau$ satisfies

$$\Phi(G) = \psi(\lambda^+(G)) \geq \psi(\nu_\tau).$$

Equality holds if and only if all nonzero eigenvalues of G are equal to $\tau/(K-1)$. Because $G\mathbf{1} = 0$, this forces

$$G = \frac{\tau}{K-1} \Pi_K = G_\tau^{\text{iso}}.$$

Hence G_τ^{iso} is the unique global minimizer.

Step 4: local minimizers and gradient-like dynamics. The set $\mathcal{G}_\tau$ is convex and compact, and Φ is convex on it because it is a convex spectral functional restricted to an affine slice of the positive semidefinite cone. Therefore every local minimizer is global, hence equal to G_τ^{iso} . If G_t is any gradient-like trajectory whose Lyapunov functional is Φ and whose limit exists and is a local minimizer, the limit must be G_τ^{iso} . $\square$

Corollary 6.15 (A concrete isotropization flow). *On $\mathcal{G}_\tau$, consider the quadratic spectral anisotropy functional*

$$\Phi_{\text{iso}}(G) := \frac{1}{2} \left\| G - \frac{\tau}{K-1} \Pi_K \right\|_F^2. \quad (85)$$

Then Φ_{iso} satisfies the hypotheses of Theorem 6.14. Moreover, its projected gradient flow on $\mathcal{G}_\tau$ is explicitly

$$\dot{G}_t = - \left(G_t - \frac{\tau}{K-1} \Pi_K \right), \quad (86)$$

so that

$$G_t = \frac{\tau}{K-1} \Pi_K + e^{-t} \left(G_0 - \frac{\tau}{K-1} \Pi_K \right). \quad (87)$$

Consequently, the normalized isotropy defect

$$\Delta_t^{\text{iso}} := \frac{1}{\tau} \left\| G_t - \frac{\tau}{K-1} \Pi_K \right\|_F \quad (88)$$

obeys the exact exponential law

$$\Delta_t^{\text{iso}} = e^{-t} \Delta_0^{\text{iso}}. \quad (89)$$

Proof sketch. The functional (85) is spectral, convex, symmetric in the nonzero eigenvalues, and strictly Schur-convex at fixed trace because it is the squared Euclidean distance from the uniform spectrum. Since $G_t - \frac{\tau}{K-1} \Pi_K$ already lies in the tangent space of the affine constraints $G\mathbf{1} = 0$ and $\text{Tr}(G) = \tau$, the projected gradient equals the ordinary gradient. Solving the linear ODE (86) yields (87) and hence (89). $\square$

Theorem 6.16 (Max–min prototype margin selects the isotropic projector). *For $G \in \mathcal{G}_\tau$, define the reduced prototype margin*

$$\mathfrak{m}(G) := \min_{1 \leq c \leq K} \min_{j \neq c} (G_{cc} - G_{jc}). \quad (90)$$

Then every $G \in \mathcal{G}_\tau$ satisfies

$$\mathfrak{m}(G) \leq \frac{\tau}{K-1}. \quad (91)$$

Moreover, equality holds if and only if

$$G = G_\tau^{\text{iso}} = \frac{\tau}{K-1} \Pi_K. \quad (92)$$

Proof sketch. Fix $c \in \{1, \dots, K\}$. Since $G\mathbf{1} = 0$, the c th column sums to zero:

$$\sum_{j \neq c} G_{jc} = -G_{cc}.$$

Hence at least one index $j \neq c$ satisfies

$$G_{jc} \geq -\frac{G_{cc}}{K-1}.$$

Therefore

$$\min_{j \neq c} (G_{cc} - G_{jc}) \leq G_{cc} + \frac{G_{cc}}{K-1} = \frac{K}{K-1} G_{cc}.$$

Taking the minimum over c and then using $\min_c G_{cc} \leq \tau/K$ gives

$$\mathbf{m}(G) \leq \frac{K}{K-1} \min_c G_{cc} \leq \frac{K}{K-1} \cdot \frac{\tau}{K} = \frac{\tau}{K-1}.$$

This proves (91). If equality holds, then every inequality above must be an equality. Hence all diagonal entries are equal to τ/K , and for each fixed c every off-diagonal entry in the c th column must equal the row average $-G_{cc}/(K-1) = -\tau/(K(K-1))$. Thus G has constant diagonal and constant off-diagonal entries, with row sums zero and trace τ , so necessarily

$$G = \frac{\tau}{K-1} \Pi_K.$$

The converse is immediate by direct computation. $\square$

Corollary 6.17 (Large-scale reduced prototype cross-entropy selects isotropy). *Assume the late representation has entered the prototype regime, and suppose the classifier is self-dually aligned with the centered class means in the reduced sense*

$$W = \alpha M, \quad \alpha > 0, \quad (93)$$

where $M = [m^1, \dots, m^K]$ is the centered class-mean matrix and $G = M^\top M \in \mathcal{G}_\tau$. Then the reduced balanced prototype cross-entropy is

$$\mathcal{L}_\alpha(G) := \frac{1}{K} \sum_{c=1}^K \log \left(1 + \sum_{j \neq c} e^{-\alpha(G_{cc} - G_{jc})} \right). \quad (94)$$

Let $G_\alpha \in \arg \min_{G \in \mathcal{G}_\tau} \mathcal{L}_\alpha(G)$. Then every accumulation point of $(G_\alpha)_{\alpha \rightarrow \infty}$ equals G_τ^{iso} . Equivalently,

$$G_\alpha \rightarrow G_\tau^{\text{iso}} \quad \text{as } \alpha \rightarrow \infty. \quad (95)$$

Proof sketch. For each class c , define

$$\mathbf{m}_c(G) := \min_{j \neq c} (G_{cc} - G_{jc}), \quad \mathbf{m}(G) = \min_c \mathbf{m}_c(G).$$

Then for each c ,

$$\log(1 + e^{-\alpha \mathbf{m}_c(G)}) \leq \log \left(1 + \sum_{j \neq c} e^{-\alpha(G_{cc} - G_{jc})} \right) \leq \log(1 + (K-1)e^{-\alpha \mathbf{m}_c(G)}).$$

Since at least one class attains $\mathbf{m}(G)$, this implies the comparison bounds

$$\frac{1}{K} \log(1 + e^{-\alpha \mathbf{m}(G)}) \leq \mathcal{L}_\alpha(G) \leq \log(1 + (K-1)e^{-\alpha \mathbf{m}(G)}). \quad (96)$$

Thus minimizing $\mathcal{L}_\alpha$ for large α is asymptotically equivalent to maximizing the reduced prototype margin $\mathbf{m}(G)$. By Theorem 6.16, the unique maximizer of $\mathbf{m}$ on $\mathcal{G}_\tau$ is G_τ^{iso} . Compactness of $\mathcal{G}_\tau$ therefore implies that every sequence of minimizers G_α converges to G_τ^{iso} as $\alpha \rightarrow \infty$. $\square$

Corollary 6.18 (A self-dual cross-entropy route to terminal neural collapse). *Suppose a balanced classifier has entered the prototype regime of Corollary 6.10, and let*

$$\widehat{G}_t := \frac{G_t}{\text{Tr}(G_t)} \in \mathcal{G}_1. \quad (97)$$

Assume that in late training there exists a scale parameter $\alpha_t \rightarrow \infty$ such that the normalized semantic dynamics asymptotically tracks minimizers of the reduced prototype cross-entropy $\mathcal{L}_{\alpha_t}$ from (94), with self-dual reduced alignment (93). Then

$$\widehat{G}_t \rightarrow \frac{1}{K-1} \Pi_K. \quad (98)$$

If, in addition, the hypotheses of Theorem 6.12 hold asymptotically and classifier-feature alignment (75) is satisfied, then the terminal representation and the classifier both converge to simplex / ETF geometry.

Proof sketch. By Corollary 6.17, the only large-scale limit compatible with reduced self-dual prototype cross-entropy is the isotropic projector. This is exactly the vanishing of the semantic-isotropy defect. Theorem 6.12 then upgrades isotropy to simplex rigidity of the centered class means, and Corollary 6.13 transfers the same geometry to the classifier. $\square$

Corollary 6.19 (Balanced late-training selection principle for terminal neural collapse). *Suppose a balanced classifier has already entered the prototype regime of Corollary 6.10, and let*

$$\widehat{G}_t := \frac{G_t}{\text{Tr}(G_t)} \in \mathcal{G}_1. \quad (99)$$

Assume the late-phase semantic dynamics of $\widehat{G}_t$ is gradient-like with Lyapunov functional satisfying the hypotheses of Theorem 6.14, and that the trajectory converges to a local minimizer. Then

$$\delta_t^{\text{iso}} \rightarrow 0. \quad (100)$$

If, in addition, the hypotheses of Theorem 6.12 hold asymptotically and classifier-feature alignment (75) is satisfied, then the terminal representation and the classifier both converge to simplex / ETF geometry.

Proof sketch. By Theorem 6.14, the only local minimizer compatible with the balanced spectral Lyapunov functional is $\Pi_K/(K-1)$. Hence $\widehat{G}_t \rightarrow \Pi_K/(K-1)$, which is exactly the vanishing of the semantic-isotropy defect. Theorem 6.12 then turns isotropy into simplex rigidity of the centered class means, and Corollary 6.13 transfers the same geometry to the classifier. $\square$

Deep-learning reading. Theorem 6.14 says that the missing ingredient behind terminal neural collapse need not be interpreted as a mysterious extra regularity. In a balanced late-training regime, once features have collapsed to class prototypes, the surviving semantic degrees of freedom are encoded by a centered Gram matrix. If the reduced training dynamics is permutation-symmetric across classes and decreases any convex strictly Schur-convex spectral anisotropy energy on that Gram matrix, then the isotropic projector is the unique stable terminal selection. Theorem 6.16 and Corollaries 6.17 and 6.18 make this more concrete: in a balanced self-dual prototype regime, large-scale reduced cross-entropy itself asymptotically selects the same isotropic projector because

it is the unique maximizer of the minimum prototype margin at fixed trace. Combined with Theorem 6.12, this yields a three-stage picture of terminal neural collapse:

$$\begin{aligned} \text{Jacobian-chain compression of class fibers} &\implies \text{prototype formation} \\ &\implies \text{balanced spectral / margin selection of isotropy} \\ &\implies \text{simplex / ETF terminal geometry.} \end{aligned}$$

In this sense, terminal neural collapse can be read not as an isolated empirical surprise, but as the natural terminal selection law for a balanced compressed semantic representation, with a reduced cross-entropy route in the self-dual late phase.

6.7 From full-network late training to reduced semantic dynamics

The reduced selection theorems identify the target geometry, but they still live on a simplified semantic state space. The next theorem explains when full-network late training inherits that reduced selector. Its role is to justify that the isotropic projector is not merely the optimizer of a toy model, but the effective selector seen by the original cross-entropy dynamics once the network has entered a balanced prototype phase.

The reduced prototype route of Corollaries 6.17 and 6.18 still leaves one important gap from the perspective of deep learning: those results live on a reduced prototype-semantic model, whereas the actual network trains in parameter space. The purpose of this subsection is to close that gap at the level of a late-phase reduction theorem.

Consider a balanced training set with n samples per class, a feature map $z_\phi(x) \in \mathbb{R}^d$, and a linear classifier $H = [h^1, \dots, h^K] \in \mathbb{R}^{d \times K}$. Write the full empirical cross-entropy as

$$\mathcal{L}_{\text{CE}}(\phi, H) := \frac{1}{nK} \sum_{i=1}^{nK} \ell(H^\top z_\phi(x_i), y_i), \quad (101)$$

where $\ell(s, y) = -s_y + \log \sum_{j=1}^K e^{s_j}$. We consider the gradient flow

$$\dot{\phi}_t = -\nabla_\phi \mathcal{L}_{\text{CE}}(\phi_t, H_t), \quad \dot{H}_t = -\nabla_H \mathcal{L}_{\text{CE}}(\phi_t, H_t). \quad (102)$$

Let

$$z_i(t) := z_{\phi_t}(x_i), \quad \mu_t^c := \frac{1}{n} \sum_{i: y_i=c} z_i(t), \quad m_t^c := \mu_t^c - \bar{\mu}_t,$$

where $\bar{\mu}_t := K^{-1} \sum_c \mu_t^c$. Let $M_t = [m_t^1, \dots, m_t^K]$, $G_t = M_t^\top M_t$, and $\hat{G}_t := G_t / \text{Tr}(G_t)$ whenever $\text{Tr}(G_t) > 0$.

We begin with the feature-level reduction from full cross-entropy gradients to the reduced prototype gradients.

Lemma 6.20 (Prototype reduction of class-averaged feature gradients). *For each class c , define the class-averaged feature gradient*

$$g_t^c := \frac{1}{n} \sum_{i: y_i=c} \nabla_{z_i} \ell(H_t^\top z_i(t), y_i). \quad (103)$$

Assume that for some $t \geq t_0$ the features are in a prototype regime with

$$z_i(t) = \mu_t^{y_i} + \delta_i(t), \quad \max_{1 \leq i \leq nK} \|\delta_i(t)\| \leq r_t, \quad (104)$$

and the classifier is approximately self-dual,

$$h_t^c = \alpha_t m_t^c + \xi_t^c, \quad \alpha_t > 0, \quad \xi_t := \max_{1 \leq c \leq K} \|\xi_t^c\|. \quad (105)$$

Assume also that the class means remain in a bounded region of feature space. Then there exists a constant $C > 0$, depending only on that bounded region, such that

$$\max_{1 \leq c \leq K} \|g_t^c - \nabla_{m^c} \mathcal{L}_{\alpha_t}(G_t)\| \leq C(\alpha_t r_t + \xi_t), \quad (106)$$

where $\mathcal{L}_{\alpha}$ is the reduced prototype cross-entropy from (94).

Proof sketch. For a sample i of class c , the feature gradient is

$$\nabla_{z_i} \ell = \sum_{j=1}^K p_{ij} h_t^j - h_t^c, \quad p_{ij} := \frac{e^{\langle h_t^j, z_i(t) \rangle}}{\sum_{k=1}^K e^{\langle h_t^k, z_i(t) \rangle}}.$$

Under (104) and (105), the logits differ from the reduced prototype logits $\alpha_t \langle m_t^j, m_t^c \rangle$ by at most $C(\alpha_t r_t + \xi_t)$. Softmax Lipschitz continuity on bounded logit sets transfers the same error scale to the probabilities p_{ij} . Averaging over all samples in class c then yields the class-averaged feature gradient approximation (106). $\square$

The next theorem pushes this reduction through the parameter dynamics. Its key object is the late-phase semantic NTK on the class means. Define the class-mean NTK blocks

$$\mathbb{K}_t^{cc'} := \frac{1}{n^2} \sum_{i: y_i=c} \sum_{j: y_j=c'} D_{\phi} z_{\phi_t}(x_i) D_{\phi} z_{\phi_t}(x_j)^{\top} \in \mathbb{R}^{d \times d}, \quad (107)$$

and let $\mathbb{K}_t = (\mathbb{K}_t^{cc'})_{1 \leq c, c' \leq K}$ be the associated block operator on the product space $(\mathbb{R}^d)^K$ endowed with the product Euclidean norm.

Theorem 6.21 (Late-phase cross-entropy gradient flow reduces to normalized prototype-semantic flow). *Consider the full gradient flow (102) on a balanced training set. Assume that for all sufficiently large t the following hold.*

(R1) Prototype concentration.

$$r_t := \max_{1 \leq i \leq nK} \|z_i(t) - \mu_t^{y_i}\| \rightarrow 0. \quad (108)$$

(R2) Self-dual alignment with diverging scale. *There exist $\alpha_t \rightarrow \infty$ and classifier errors ξ_t^c such that*

$$h_t^c = \alpha_t m_t^c + \xi_t^c, \quad \xi_t := \max_{1 \leq c \leq K} \|\xi_t^c\|, \quad \frac{\xi_t}{\alpha_t} \rightarrow 0. \quad (109)$$

(R3) Semantic-scale nondegeneracy. *There exist constants $0 < \tau_- \leq \tau_+ < \infty$ such that*

$$\tau_- \leq \text{Tr}(G_t) \leq \tau_+. \quad (110)$$

(R4) Class-balanced semantic NTK rigidity. *Writing $E_t := \text{span}\{m_t^1, \dots, m_t^K\}$ and $E_t^K := E_t \times \dots \times E_t$, there exists a scalar $\kappa_t > 0$ such that*

$$\Delta_t^{\text{sNTK}} := \frac{1}{\kappa_t} \left\| \mathbb{K}_t|_{E_t^K} - \kappa_t I_{E_t^K} \right\|_{\text{op}} \rightarrow 0, \quad \int_{t_0}^{\infty} \kappa_t dt = \infty. \quad (111)$$

(R5) Scale separation.

$$\frac{\alpha_t r_t + \xi_t}{\kappa_t} \rightarrow 0. \quad (112)$$

Then there exists a remainder matrix $R_t \in T_{\hat{G}_t} \mathcal{G}_1$ such that, after the time reparametrization

$$s(t) := \int_{t_0}^t 2\kappa_u \operatorname{Tr}(G_u) du, \quad (113)$$

one has

$$\frac{d}{ds} \hat{G}_t = -\operatorname{grad}_{\mathcal{G}_1} \mathcal{L}_{\alpha_t}(\hat{G}_t) + R_t, \quad (114)$$

with the estimate

$$\|R_t\|_F \leq C \left(\Delta_t^{\text{sNTK}} + \frac{\alpha_t r_t + \xi_t}{\kappa_t} \right). \quad (115)$$

In particular,

$$\left\| \frac{d}{ds} \hat{G}_t + \operatorname{grad}_{\mathcal{G}_1} \mathcal{L}_{\alpha_t}(\hat{G}_t) \right\|_F \rightarrow 0, \quad (116)$$

so the normalized semantic Gram trajectory is an asymptotic pseudotrajectory of the reduced prototype-semantic flow.

If, moreover, the limiting reduced flow on $\mathcal{G}_1$ is gradient-like with unique asymptotically stable equilibrium

$$G_1^{\text{iso}} := \frac{1}{K-1} \Pi_K, \quad (117)$$

as in Corollary 6.17, then

$$\hat{G}_t \rightarrow G_1^{\text{iso}}. \quad (118)$$

If, in addition, the hypotheses of Theorem 6.12 and Corollary 6.13 hold asymptotically, then the full network enters terminal neural collapse.

Proof sketch. The proof has three steps.

Step 1: exact mean dynamics under parameter gradient flow. Differentiating each class mean along (102) and using the chain rule gives

$$\dot{\mu}_t^c = - \sum_{c'=1}^K \mathbf{K}_t^{cc'} g_t^{c'},$$

where $g_t^{c'}$ is the class-averaged feature gradient from (103). Passing to centered means eliminates the trivial center-of-mass direction and yields the same equation on E_t^K .

Step 2: reduction to prototype gradients. By Lemma 6.20,

$$g_t^c = \nabla_{m^c} \mathcal{L}_{\alpha_t}(G_t) + \varepsilon_t^c, \quad \max_c \|\varepsilon_t^c\| \leq C(\alpha_t r_t + \xi_t).$$

Assumption (111) then implies

$$\dot{m}_t^c = -\kappa_t \nabla_{m^c} \mathcal{L}_{\alpha_t}(G_t) + \tilde{\varepsilon}_t^c,$$

with

$$\max_c \|\tilde{\varepsilon}_t^c\| \leq C(\kappa_t \Delta_t^{\text{sNTK}} + \alpha_t r_t + \xi_t).$$

Step 3: induced Gram dynamics and normalization. Differentiating $G_t = M_t^\top M_t$ yields

$$\dot{G}_t = -2\kappa_t \operatorname{grad}_{\mathcal{G}_{\operatorname{Tr}(G_t)}} \mathcal{L}_{\alpha_t}(G_t) + \mathcal{E}_t,$$

where $\|\mathcal{E}_t\|_F \leq C \operatorname{Tr}(G_t)(\kappa_t \Delta_t^{\text{NTK}} + \alpha_t r_t + \xi_t)$. Dividing by $\operatorname{Tr}(G_t)$ and subtracting the scalar radial component gives the normalized equation (114). The bound (115) follows from (110) and (112). The asymptotic-pseudotrajectory claim then follows immediately, and any unique asymptotically stable selector of the reduced flow is inherited by the full trajectory. $\square$

Corollary 6.22 (Small-step SGD inherits the same semantic selector). *Consider mini-batch SGD applied to the same balanced network with step sizes $(\eta_k)_{k \geq 0}$ satisfying*

$$\sum_{k=0}^{\infty} \eta_k = \infty, \quad \sum_{k=0}^{\infty} \eta_k^2 < \infty, \quad (119)$$

and unbiased gradient noise with uniformly bounded conditional second moments. Assume the discrete late-phase analogues of (R1)–(R5) hold along the iterates, and let $\bar{G}(s)$ be the piecewise-linear interpolation of the normalized semantic Gram matrices $\hat{G}_k$. Then $\bar{G}(s)$ is an asymptotic pseudotrajectory of the same reduced prototype-semantic flow from Theorem 6.21. In particular, if the reduced flow has the unique asymptotically stable selector G_1^{iso} , then along every almost surely convergent run,

$$\hat{G}_k \rightarrow G_1^{\text{iso}} = \frac{1}{K-1} \Pi_K \quad \text{almost surely.} \quad (120)$$

Under the additional hypotheses of Theorem 6.12 and Corollary 6.13, the same terminal-neural-collapse conclusion holds almost surely.

Proof sketch. The SGD iterate on the semantic Gram inherits the same deterministic drift as in Theorem 6.21, with an additional martingale-difference noise term of order η_k . The assumptions (119) imply that the cumulative noise is asymptotically negligible on finite windows, so the interpolated trajectory is an asymptotic pseudotrajectory of the reduced ODE. Any unique asymptotically stable selector of that ODE is therefore inherited by the stochastic iterates. The final implication to terminal neural collapse is the same as in Theorem 6.21. $\square$

Deep-learning reading. Theorem 6.21 says that the late balanced prototype phase of full-network training has an unexpectedly rigid structure. Once within-class variability is already small, once the classifier has largely aligned with the class prototypes, and once the semantic NTK seen by the class means becomes approximately class-balanced, the original parameter-space cross-entropy gradient flow can no longer move the semantic representation arbitrarily. Up to vanishing error, it is forced to follow the reduced prototype-semantic dynamics on the normalized class-mean Gram matrix. In that regime, the selector identified earlier on the reduced space is not merely an artifact of a toy model: it becomes the effective selector of the original network as well. Corollary 6.22 says the same mechanism persists for small-step SGD. From a deep-learning perspective, this turns terminal neural collapse into a late-phase training law: balanced optimization first creates class prototypes, then compresses the residual semantic degrees of freedom into the unique isotropic selector, and finally rigidifies them into simplex / ETF geometry.

6.8 A direct full-flow route to semantic isotropy

The reduction theorem still routes isotropy through a reduced semantic model. The next result moves one step closer to a direct theory of training by replacing the external reduced selector with an explicit anisotropy-dissipation condition measured along the full late-phase semantic trajectory itself.

Theorem 6.21 reduces the full late-phase parameter dynamics to a normalized prototype-semantic flow and shows that any unique stable selector of the reduced system is inherited by the original network. The remaining conceptual gap is that isotropy selection still enters through the reduced model. The next theorem removes that extra selector hypothesis and replaces it by a direct quantitative dissipation condition along the full late-phase semantic trajectory itself.

Define the isotropy energy on $\mathcal{G}_1$ by

$$\mathcal{E}_{\text{iso}}(G) := \frac{1}{2} \left\| G - \frac{1}{K-1} \Pi_K \right\|_F^2. \quad (121)$$

Whenever $\mathcal{E}_{\text{iso}}(\hat{G}_t) > 0$, define the full-flow semantic anisotropy coercivity ratio

$$\mathfrak{c}_t^{\text{iso}} := \frac{\langle \text{grad}_{\mathcal{G}_1} \mathcal{E}_{\text{iso}}(\hat{G}_t), \text{grad}_{\mathcal{G}_1} \mathcal{L}_{\alpha_t}(\hat{G}_t) \rangle_F}{\mathcal{E}_{\text{iso}}(\hat{G}_t)}. \quad (122)$$

This quantity is experimentally accessible: it asks whether the late semantic drift seen by the full network really points down the isotropy energy at a rate proportional to the current anisotropy level.

Theorem 6.23 (Direct full-flow anisotropy dissipation selects semantic isotropy). *Assume the hypotheses of Theorem 6.21. In addition assume that there exist $s_0 < \infty$ and $\lambda_* > 0$ such that for all reduced times $s \geq s_0$ with $\mathcal{E}_{\text{iso}}(\hat{G}_t) > 0$,*

$$\mathfrak{c}_t^{\text{iso}} \geq \lambda_*. \quad (123)$$

Assume moreover that the reduction remainder from (114) is square integrable in reduced time:

$$\int_{s_0}^{\infty} \|R_t\|_F^2 ds < \infty. \quad (124)$$

Then for all $s \geq s_0$,

$$\mathcal{E}_{\text{iso}}(\hat{G}_t) \leq \exp\left(-\frac{\lambda_*}{2}(s - s_0)\right) \mathcal{E}_{\text{iso}}(\hat{G}_{t_0}) + \frac{1}{\lambda_*} \int_{s_0}^s \exp\left(-\frac{\lambda_*}{2}(s - u)\right) \|R_u\|_F^2 du. \quad (125)$$

In particular,

$$\mathcal{E}_{\text{iso}}(\hat{G}_t) \rightarrow 0, \quad \delta_t^{\text{iso}} \rightarrow 0, \quad \hat{G}_t \rightarrow \frac{1}{K-1} \Pi_K. \quad (126)$$

A sufficient condition for (124) is

$$\int_{s_0}^{\infty} \left(\Delta_t^{\text{sNTK}}\right)^2 ds < \infty, \quad \int_{s_0}^{\infty} \left(\frac{\alpha_t r_t + \xi_t}{\kappa_t}\right)^2 ds < \infty, \quad (127)$$

by the bound (115).

Proof sketch. Differentiate (121) along the reduced-time flow (114). Since the difference of two points of $\mathcal{G}_1$ is centered, symmetric, and trace-free, one has

$$\text{grad}_{\mathcal{G}_1} \mathcal{E}_{\text{iso}}(\hat{G}_t) = \hat{G}_t - \frac{1}{K-1} \Pi_K.$$

Therefore

$$\frac{d}{ds} \mathcal{E}_{\text{iso}}(\hat{G}_t) = - \left\langle \text{grad}_{\mathcal{G}_1} \mathcal{E}_{\text{iso}}(\hat{G}_t), \text{grad}_{\mathcal{G}_1} \mathcal{L}_{\alpha_t}(\hat{G}_t) \right\rangle_F + \left\langle \text{grad}_{\mathcal{G}_1} \mathcal{E}_{\text{iso}}(\hat{G}_t), R_t \right\rangle_F.$$

Assumption (123) gives the coercive term

$$- \left\langle \text{grad} \mathcal{E}_{\text{iso}}, \text{grad} \mathcal{L}_{\alpha_t} \right\rangle_F \leq -\lambda_* \mathcal{E}_{\text{iso}}(\hat{G}_t).$$

Using $\|\text{grad} \mathcal{E}_{\text{iso}}\|_F = \sqrt{2\mathcal{E}_{\text{iso}}}$ together with Young's inequality,

$$\sqrt{2\mathcal{E}_{\text{iso}}} \|R_t\|_F \leq \frac{\lambda_*}{2} \mathcal{E}_{\text{iso}}(\hat{G}_t) + \frac{1}{\lambda_*} \|R_t\|_F^2.$$

Hence

$$\frac{d}{ds} \mathcal{E}_{\text{iso}}(\hat{G}_t) \leq -\frac{\lambda_*}{2} \mathcal{E}_{\text{iso}}(\hat{G}_t) + \frac{1}{\lambda_*} \|R_t\|_F^2.$$

Gronwall's inequality yields (125). The convergence (126) follows from (124). The sufficient condition (127) is immediate from (115). $\square$

Corollary 6.24 (Full-flow saturation of the normalized prototype margin). *Assume the hypotheses of Theorem 6.23. Define the normalized prototype margin*

$$\hat{m}_t := \min_{c \neq c'} \frac{\|m_t^c - m_t^{c'}\|^2}{\text{Tr}(G_t)}. \quad (128)$$

Then

$$\hat{m}_t \rightarrow \frac{2}{K-1}. \quad (129)$$

Proof sketch. By (126),

$$\frac{G_t}{\text{Tr}(G_t)} = \hat{G}_t \rightarrow \frac{1}{K-1} \Pi_K.$$

The diagonal entries of the limit are $1/K$, while the off-diagonal entries are $-1/(K(K-1))$. Therefore for every $c \neq c'$,

$$\frac{\|m_t^c - m_t^{c'}\|^2}{\text{Tr}(G_t)} = \frac{G_t^{cc} + G_t^{c'c'} - 2G_t^{cc'}}{\text{Tr}(G_t)} \rightarrow \frac{2}{K-1},$$

and the minimum over finitely many pairs has the same limit. $\square$

Corollary 6.25 (Direct full-flow route to terminal neural collapse). *Assume the hypotheses of Theorem 6.23. Suppose in addition that the asymptotic full-rank and trace hypotheses from Theorem 6.12 hold, and that the prototype drift satisfies $\nu_t \rightarrow 0$. Then the centered class means converge to regular-simplex / ETF geometry. If, moreover, the primitive regularized head flow (134) is run on the evolving prototypes and the hypotheses of Theorem 6.26 hold, then*

$$\mathcal{A}_t^{\text{head}} \rightarrow 0, \quad (130)$$

and the joint feature-classifier geometry converges to self-dual simplex / ETF terminal neural collapse.

Proof sketch. Theorem 6.23 gives the isotropy hypothesis required by Theorem 6.12, so the class means converge to regular-simplex geometry. The additional prototype-drift hypothesis $\nu_t \rightarrow 0$ together with isotropy now places the primitive head flow exactly in the regime of Theorem 6.26, which yields $\mathcal{A}_t^{\text{head}} \rightarrow 0$. The self-dual simplex / ETF conclusion follows exactly as in Corollary 6.27. $\square$

Deep-learning reading. Theorem 6.23 upgrades the reduced-selector picture to a directly testable full-network statement. It says that once the late semantic representation already behaves like a compressed prototype system, the remaining question is whether the full semantic drift dissipates anisotropy faster than the reduction errors accumulate. If it does, then isotropy is not an external hypothesis and not merely a property of a toy reduced model: it is a quantitative consequence of the end-to-end late-phase dynamics itself. Corollary 6.24 shows what deep-learning practitioners should look for: the normalized prototype margin should saturate to the regular-simplex value $2/(K-1)$. Corollary 6.25 then closes the loop to terminal neural collapse. In plain terms, late training first creates prototypes, then directly equalizes the surviving semantic directions, and only then locks the classifier onto the same simplex geometry.

6.9 Endogenous classifier-feature alignment

The final visible piece of terminal neural collapse is self-duality: classifier directions align with the same simplex geometry carried by the class means. Rather than treating that alignment as an external closure assumption, the next theorem shows that a primitive regularized head dynamics can generate it once semantic isotropy has already been selected and prototype drift has slowed.

After Theorem 6.23, the selector of semantic isotropy is no longer external. The remaining visible gap is the last classifier-feature alignment step. The purpose of the present subsection is to internalize that final stage. We show that once the semantic representation has already selected isotropy and the class prototypes move slowly, a primitive ℓ_2 -regularized cross-entropy head flow automatically captures the semantic span and converges toward the self-dual ray.

Fix a time-dependent centered prototype matrix

$$M_t = [m_t^1, \dots, m_t^K] \in \mathbb{R}^{d \times K}, \quad \sum_{c=1}^K m_t^c = 0,$$

with semantic span $E_t := \text{span}\{m_t^1, \dots, m_t^K\}$, projector $P_t := P_{E_t}$, and normalized centered Gram

$$\hat{G}_t := \frac{M_t^\top M_t}{\text{Tr}(M_t^\top M_t)} \in \mathcal{G}_1.$$

Define the semantic-isotropy defect

$$\delta_t^{\text{iso}} := \left\| \hat{G}_t - \frac{1}{K-1} \Pi_K \right\|_F, \quad (131)$$

and the prototype drift rate

$$\nu_t := \left\| \dot{M}_t \right\|_F. \quad (132)$$

For a classifier head $H = [h^1, \dots, h^K] \in \mathbb{R}^{d \times K}$, consider the primitive balanced prototype-head loss with explicit ℓ_2 regularization:

$$\mathcal{L}_{\text{head}, \lambda}(H; M_t) := \frac{1}{K} \sum_{c=1}^K \left(-\langle h^c, m_t^c \rangle + \log \sum_{j=1}^K e^{\langle h^j, m_t^c \rangle} \right) + \frac{\lambda}{2} \|H\|_F^2, \quad \lambda > 0. \quad (133)$$

We study the time-dependent head gradient flow

$$\dot{H}_t = -\nabla_H \mathcal{L}_{\text{head}, \lambda}(H_t; M_t). \quad (134)$$

The key quantity is the primitive self-dual alignment defect

$$\mathcal{A}_t^{\text{head}} := \inf_{\alpha \geq 0} \|H_t - \alpha M_t\|_F. \quad (135)$$

Theorem 6.26 (Primitive regularized head dynamics selects the self-dual ray after isotropy). *Consider the time-dependent head flow (134). Assume:*

(H1) Balanced semantic scale. *There exist constants $0 < \tau_- \leq \tau_+ < \infty$ such that*

$$\tau_- \leq \text{Tr}(M_t^\top M_t) \leq \tau_+. \quad (136)$$

(H2) Bounded prototype regime. *There exists $M_+ < \infty$ such that*

$$\|M_t\|_F \leq M_+ \quad \text{for all } t \geq t_0. \quad (137)$$

(H3) Slow prototype drift. *The map $t \mapsto M_t$ is C^1 for $t \geq t_0$, and*

$$\nu_t = \left\| \dot{M}_t \right\|_F \rightarrow 0. \quad (138)$$

Then, for every $t \geq t_0$, the following hold.

(a) Unique instantaneous minimizer and semantic-span capture. *The map $H \mapsto \mathcal{L}_{\text{head}, \lambda}(H; M_t)$ is λ -strongly convex, so it admits a unique minimizer H_t^* . Moreover,*

$$(I - P_t)H_t^* = 0. \quad (139)$$

(b) Exact isotropy implies an exact self-dual minimizer. *If*

$$\delta_t^{\text{iso}} = 0, \quad (140)$$

then there exists a unique scalar $\alpha_t^ > 0$ such that*

$$H_t^* = \alpha_t^* M_t. \quad (141)$$

(c) Near-isotropy implies near-self-dual minimizers. *There exist constants $\bar{\delta} > 0$ and $C_{\text{head}} > 0$, depending only on $\lambda, \tau_-, \tau_+, M_+$, and K , such that whenever $\delta_t^{\text{iso}} \leq \bar{\delta}$,*

$$\inf_{\alpha \geq 0} \|H_t^* - \alpha M_t\|_F \leq C_{\text{head}} \delta_t^{\text{iso}}. \quad (142)$$

(d) Exponential tracking of the instantaneous minimizer. *There exists $C_{\text{trk}} > 0$ such that*

$$\|H_t - H_t^*\|_F \leq e^{-\lambda(t-t_0)} \|H_{t_0} - H_{t_0}^*\|_F + C_{\text{trk}} \int_{t_0}^t e^{-\lambda(t-s)} \nu_s ds. \quad (143)$$

Consequently, whenever $\delta_t^{\text{iso}} \rightarrow 0$ and $\nu_t \rightarrow 0$,

$$\mathcal{A}_t^{\text{head}} \leq e^{-\lambda(t-t_0)} \|H_{t_0} - H_{t_0}^*\|_F + C_{\text{trk}} \int_{t_0}^t e^{-\lambda(t-s)} \nu_s ds + C_{\text{head}} \delta_t^{\text{iso}} \rightarrow 0. \quad (144)$$

In words: after isotropy has been selected and prototype drift has slowed, the primitive regularized head dynamics forces the classifier onto the self-dual ray.

Proof sketch. The proof has four steps.

Step 1: strong convexity and semantic-span capture. For fixed M_t , the loss (133) depends on H through the prototype logits $M_t^\top H$, plus the strictly convex term $(\lambda/2) \|H\|_F^2$. Hence the objective is λ -strongly convex and has a unique minimizer. If $Q_t := I - P_t$, then replacing H by $P_t H$ leaves all prototype logits unchanged while strictly reducing the regularizer unless $Q_t H = 0$. Therefore any minimizer must satisfy (139).

Step 2: exact isotropy gives an exact self-dual minimizer. Under $\delta_t^{\text{iso}} = 0$, the centered prototype Gram is exactly

$$M_t^\top M_t = \frac{\text{Tr}(M_t^\top M_t)}{K-1} \Pi_K.$$

After Step 1, write $H_t^* = M_t A_t^*$ on the semantic span. The reduced objective is invariant under simultaneous permutations of class labels because both the prototype configuration and the balanced head loss are permutation-symmetric. Since the minimizer is unique, A_t^* must commute with the permutation representation on the centered class subspace; by the commutant structure of the permutation action, A_t^* must be of the form $a_t I + b_t \mathbf{1}\mathbf{1}^\top$. Because $M_t \mathbf{1} = 0$, the second term vanishes, so $H_t^* = a_t M_t$. Positivity of $a_t = \alpha_t^*$ follows from the fact that cross-entropy favors correct-class logits.

Step 3: near-isotropy gives a near-self-dual minimizer. The map $(H, M) \mapsto \nabla_H \mathcal{L}_{\text{head}, \lambda}(H; M)$ is smooth on bounded sets, and the Hessian in H is bounded below by λI . Hence the implicit minimizer map $M \mapsto H^*(M)$ is locally Lipschitz on the bounded semantic-scale set from (136)–(137). Perturbing from an exactly isotropic prototype configuration to one with defect δ_t^{iso} gives (142).

Step 4: tracking the time-dependent minimizer. For the time-dependent flow (134), strong convexity yields the differential inequality

$$\frac{d}{dt} \|H_t - H_t^*\|_F \leq -\lambda \|H_t - H_t^*\|_F + \|\dot{H}_t^*\|_F.$$

The Lipschitz dependence of $H^*(M)$ on M implies $\|\dot{H}_t^*\|_F \leq C_{\text{trk}} \nu_t$. Grönwall's inequality gives (143). Combining this tracking estimate with (142) yields (144). $\square$

Corollary 6.27 (Terminal self-duality from semantic isotropy plus primitive head dynamics). *Assume the hypotheses of Theorem 6.21, and suppose the reduced selector is inherited so that*

$$\hat{G}_t \rightarrow \frac{1}{K-1} \Pi_K.$$

Assume further that after some late time t_0 the classifier head follows the primitive regularized flow (134) with $\lambda > 0$ and the prototype drift satisfies $\nu_t \rightarrow 0$. Then

$$\mathcal{A}_t^{\text{head}} = \inf_{\alpha \geq 0} \|H_t - \alpha M_t\|_F \rightarrow 0. \quad (145)$$

Consequently, the explicit classifier–feature alignment hypothesis in Corollary 6.13 can be replaced by the primitive regularized head-dynamics assumption above, and the same self-dual simplex / ETF conclusion follows.

Proof sketch. Theorem 6.21 identifies the reduced semantic selector on the normalized Gram space and, under the selector hypothesis, gives $\hat{G}_t \rightarrow (K-1)^{-1}\Pi_K$. Theorem 6.26 then applies directly: isotropy plus slow prototype drift imply $\mathcal{A}_t^{\text{head}} \rightarrow 0$. Once the classifier head is asymptotically self-dual with the centered class means, Theorem 6.12 and Corollary 6.13 upgrade the isotropic semantic limit to the full self-dual simplex / ETF terminal geometry. $\square$

Deep-learning reading. Theorem 6.26 says that the final stage of terminal neural collapse need not be imposed by hand. Once the semantic representation has already compressed into balanced class prototypes and the normalized prototype Gram has become nearly isotropic, a primitive head-only cross-entropy dynamics with explicit weight decay has no freedom left except to lock onto the self-dual ray. The classifier first sheds any component orthogonal to the semantic span because such directions do not help the logits but do increase the regularizer, and then the in-span dynamics converges to the unique permutation-symmetric minimizer, namely the self-dual classifier. Corollary 6.27 therefore turns the last alignment step of terminal neural collapse into an endogenous training consequence rather than an external alignment axiom.

6.10 Dependency map and proof architecture

The hard theorem package is easiest to read as a dependency graph rather than as a linear list. This subsection records that graph explicitly: which lemmas feed the bridge theorem, which reduced-selection statements feed isotropy and rigidity, and where the full-flow and head-dynamics results close the loop back to deep learning.

The proof of Theorem 6.6 is built from four supporting lemmas. The main point is that the previous Jacobian-propagation and curvature-accumulation statements are no longer the headline results; they now serve as technical inputs into a representation-level theorem.

Lemma 6.28 (From local semantic cones to class-mean secants). *There exists a universal constant $C_{\text{sec}} > 0$ such that, for every pair of distinct classes $c \neq c'$, one has*

$$(1 - \delta_l - C_{\text{sec}}\chi_l r_l) \left\| \mu_l^c - \mu_l^{c'} \right\| \leq \left\| \mu_{l+1}^c - \mu_{l+1}^{c'} \right\| \leq (1 + \delta_l + C_{\text{sec}}\chi_l r_l) \left\| \mu_l^c - \mu_l^{c'} \right\|. \quad (146)$$

In particular, if $\varepsilon_l := \delta_l + C_{\text{sec}}\chi_l r_l < 1$, then semantic secants between class means remain uniformly nondegenerate up to the multiplicative factors $(1 \pm \varepsilon_l)$.

Proof sketch. Fix $z \in M_l^c$ and expand Φ_l at z in the direction of a vector $v \in \mathcal{C}_l^{\text{sem}}(z)$. The Jacobian part is controlled by the local semantic near-isometry assumption, while the Taylor remainder is bounded by $\chi_l \|v\|^2$. Averaging the resulting estimate around the class means and using the class radius bound r_l yields (146). $\square$

Lemma 6.29 (One-step transport of class moments). *For each class c , write*

$$Z_l = \mu_l^c + \xi_l^c, \quad \mathbb{E}[\xi_l^c \mid Y = c] = 0.$$

Then

$$\Phi_l(Z_l) = \Phi_l(\mu_l^c) + J_l(\mu_l^c)\xi_l^c + R_l^c, \quad \|R_l^c\| \leq \frac{1}{2}\chi_l \|\xi_l^c\|^2. \quad (147)$$

Consequently, there is a matrix error term E_l^c such that

$$\Sigma_{l+1}^c = J_l(\mu_l^c) \Sigma_l^c J_l(\mu_l^c)^\top + E_l^c, \quad (148)$$

and

$$\|E_l^c\|_{\text{op}} \leq C_{\text{mom}}(J_l \chi_l r_l^3 + \chi_l^2 r_l^4) \quad (149)$$

for a universal constant $C_{\text{mom}} > 0$.

Proof sketch. Apply a second-order Taylor expansion at μ_l^c and use the fourth-moment radius r_l to bound the mixed linear-quadratic and quadratic-quadratic remainder terms. The dominant error comes from terms of size $J_l \chi_l r_l^3$ and $\chi_l^2 r_l^4$. $\square$

Lemma 6.30 (Fiber contraction implies within-class contraction). *Under the assumptions of Theorem 6.6,*

$$\text{Tr}(W_{l+1}) \leq \rho_l^2 \text{Tr}(W_l) + \eta_l. \quad (150)$$

Proof sketch. By Lemma 6.29, each class covariance is transported by the local Jacobian plus an error term. Since the Jacobian contracts every fiber tangent direction by at most ρ_l , the principal covariance contribution contracts by ρ_l^2 . Averaging over classes and absorbing the remainder into η_l gives (150). $\square$

Lemma 6.31 (Semantic secant persistence implies between-class stability). *Under the assumptions of Theorem 6.6,*

$$\lambda_{\min}^+(B_{l+1}) \geq (1 - \varepsilon_l)^2 \lambda_{\min}^+(B_l) - \eta_l. \quad (151)$$

Proof sketch. By Lemma 6.28, all semantic secants between class means are distorted by at most a factor $(1 \pm \varepsilon_l)$. Passing from pairwise secant bounds to the covariance matrix B_l yields multiplicative control on its nonzero spectrum. The same curvature floor that appears in Lemma 6.29 contributes the additive error term η_l . $\square$

Lemma 6.32 (Discrete Grönwall and collapse-ratio decay). *Suppose the recursions*

$$\text{Tr}(W_{l+1}) \leq \rho_l^2 \text{Tr}(W_l) + \eta_l, \quad \lambda_{\min}^+(B_{l+1}) \geq (1 - \varepsilon_l)^2 \lambda_{\min}^+(B_l) - \eta_l$$

hold, together with

$$\sup_l \rho_l \leq \bar{\rho} < 1, \quad \sum_l \varepsilon_l < \infty, \quad \sum_l \eta_l < \left(\prod_{l=0}^{\infty} (1 - \varepsilon_l)^2 \right) \lambda_{\min}^+(B_0).$$

Then

$$\text{Tr}(W_l) \rightarrow 0, \quad \inf_l \lambda_{\min}^+(B_l) > 0, \quad \text{CR}_l \rightarrow 0.$$

Proof sketch. Iterating the first recursion gives

$$\text{Tr}(W_l) \leq \bar{\rho}^{2l} \text{Tr}(W_0) + \sum_{s=0}^{l-1} \bar{\rho}^{2(l-1-s)} \eta_s,$$

which tends to zero because $\bar{\rho} < 1$ and $(\eta_l) \in \ell^1$. Iterating the second recursion yields

$$\lambda_{\min}^+(B_l) \geq \left(\prod_{t=0}^{l-1} (1 - \varepsilon_t)^2 \right) \lambda_{\min}^+(B_0) - \sum_{s=0}^{l-1} \eta_s,$$

which stays uniformly positive by hypothesis. Dividing the two bounds proves the decay of the collapse ratio. $\square$

Remark 6.33 (What has changed relative to a weaker theorem section). *The point of this section is structural. The earlier Jacobian propagation and curvature accumulation formulas are still useful, but they are no longer the main theorem. They have been demoted to supporting lemmas inside a stronger statement whose conclusion is a nontrivial deep-learning phenomenon. In other words, the paper now contains a genuine bridge theorem:*

$$\text{local differential geometry} \implies \text{global class statistics} \implies \text{intermediate collapse.}$$

That is the intended hard mathematical core of the framework.

7 Beyond the Hard Core: Extended Jacobian-Chain Program

The previous section contained the paper’s hard theorem package. The present section widens the aperture again. Its statements matter, but they are no longer carrying the central explanatory burden. Instead they describe the broader research program surrounding the hard core: propagation bounds, curvature bookkeeping, topology-change criteria, quotient-information optimization geometry, and architecture-level conjectures for Transformers and structure-preserving networks.

7.1 Supporting propositions beyond the hard core

Proposition 7.1 (Restricted propagation bound). *Assume each layer is a C^1 residual block*

$$\Phi_l(z) = z + h_l f_l(z), \quad A_l = Df_l(z_l),$$

and let $E_l \subset T_{z_l} M_l$ be a family of subspaces propagated along depth. If

$$h_l \left\| A_l|_{E_l} \right\|_{\text{op}} < 1 \quad \text{for all } l,$$

then

$$\prod_{i=0}^{l-1} \left(1 - h_i \left\| A_i|_{E_i} \right\|_{\text{op}} \right) \leq \sigma_{\min}(\mathcal{J}_{l:0}|_{E_0}) \leq \sigma_{\max}(\mathcal{J}_{l:0}|_{E_0}) \leq \prod_{i=0}^{l-1} \left(1 + h_i \left\| A_i|_{E_i} \right\|_{\text{op}} \right). \quad (152)$$

In the continuous-depth limit

$$\dot{z} = f(z, t), \quad \dot{J} = A(t)J,$$

one has

$$e^{\int_0^T \lambda_{\min}(S_t|_{E_t}) dt} \leq \left\| \mathcal{J}_T|_{E_0} \right\|_{\text{op}} \leq e^{\int_0^T \lambda_{\max}(S_t|_{E_t}) dt}. \quad (153)$$

Interpretation. The first-order control variable for trainability is the strain S_t , not curvature in the broad informal sense. The target geometry is therefore not global isometry, but near-isometry on the task bundle and controlled contraction on the nuisance bundle.

Proposition 7.2 (Curvature accumulation). *Let $\Phi_{L:0} = \Phi_{L-1} \circ \dots \circ \Phi_0$ and assume each layer is C^2 . Then for any $u, v \in T_x M_0$,*

$$D^2 \Phi_{L:0}(x)[u, v] = \sum_{l=0}^{L-1} \mathcal{J}_{L:l+1}(x_l) D^2 \Phi_l(x_l) [\mathcal{J}_{l:0}(x)u, \mathcal{J}_{l:0}(x)v]. \quad (154)$$

Consequently,

$$\left\| D^2\Phi_{L:0}(x)[u, v] \right\| \leq \sum_{l=0}^{L-1} \left\| \mathcal{J}_{L:l+1}(x_l) \right\|_{\text{op}} \chi_l(x_l) \left\| \mathcal{J}_{l:0}(x)u \right\| \left\| \mathcal{J}_{l:0}(x)v \right\|. \quad (155)$$

In particular, on the task bundle,

$$\left\| D^2\Phi_{L:0} \Big|_{T^{\text{task}} \times T^{\text{task}}} \right\|_{\text{op}} \leq \mathfrak{C}_L. \quad (156)$$

Interpretation. This proposition formalizes what it means for nonlinearity to become too large. Pathology is not generated solely by a layer being highly curved in isolation; it arises when second-order bending is amplified by the surrounding Jacobian chain.

Proposition 7.3 (Topology-change criterion). *Fix a class manifold M_l^c . Suppose the map $\Phi_l|_{M_l^c}$ is a proper injective immersion, hence an embedding. Then*

$$M_{l+1}^c = \Phi_l(M_l^c)$$

is diffeomorphic to M_l^c . Therefore, any layerwise change in Betti numbers, connectedness, or persistent topology must be accompanied by at least one of the following:

- (1) *rank loss of $D\Phi_l|_{TM_l^c}$,*
- (2) *loss of injectivity of $\Phi_l|_{M_l^c}$,*
- (3) *an explicit quotienting operation such as pooling, masking, collapse, or another many-to-one architectural mechanism.*

Interpretation. This criterion turns the slogan “holes disappear” into a concrete Jacobian-singularity statement. Topological change is not mysterious; it is a consequence of singular or quotient-like events in the Jacobian chain.

Proposition 7.4 (Local stability in quotient-information geometry). *Suppose optimization is carried out along the horizontal directions of the quotient space $\tilde{\Theta} = \Theta/G$, equipped with a local metric tensor G_θ (Euclidean, path geometry, Fisher geometry, or another positive-definite choice). Let Π_θ^{hor} denote the projection onto the horizontal space and let $\theta_\star$ be a critical point. Then the update*

$$\theta_{t+1} = \theta_t - \eta \Pi_{\theta_t}^{\text{hor}} G_{\theta_t}^{-1} \nabla L(\theta_t) \quad (157)$$

linearizes near $\theta_\star$ as

$$\delta_{t+1} = \left(I - \eta G_{\theta_\star}^{-1} H_{\theta_\star}^{\text{hor}} \right) \delta_t + o(\|\delta_t\|), \quad (158)$$

where $H_{\theta_\star}^{\text{hor}}$ is the Hessian restricted to the horizontal space. If $H_{\theta_\star}^{\text{hor}}$ is self-adjoint with respect to the $G_{\theta_\star}$ -inner product, then a sufficient condition for local linear stability is

$$0 < \eta < \frac{2}{\lambda_{\max}\left(G_{\theta_\star}^{-1/2} H_{\theta_\star}^{\text{hor}} G_{\theta_\star}^{-1/2}\right)}. \quad (159)$$

Interpretation. The learning rate is a geometric step size, not a purely Euclidean scalar. Even under dynamical isometry, the Fisher geometry of parameter space may still deteriorate with depth [8]. Likewise, Euclidean sharpness can be misleading when symmetry directions dominate curvature measurements [5, 11].

7.2 Architecture-level conjectures

Conjecture 7.5 (Selective-isometry-to-collapse cascade). *Suppose that in the late phase of training the network satisfies, over a substantial depth range,*

$$\alpha_l \asymp 1, \quad \sup_l \kappa_l^{\text{task}} < \infty, \quad \beta_l \rightarrow 0, \quad \mathfrak{C}_l \leq C_\star$$

for a task-dependent threshold $C_\star$. Then the network exhibits the following cascade:

- (i) within-class covariance begins to decay in intermediate layers,*
- (ii) the fraction of total variance explained by between-class covariance increases in deeper layers,*
- (iii) the final layer approaches terminal neural collapse and a simplex-ETF-type geometry.*

Interpretation. Neural collapse is not viewed here as an isolated terminal phenomenon, but as the endpoint of a deeper geometric mechanism: near-isometry on task directions plus contraction on nuisance directions drives the feature geometry toward classwise collapse [17, 19, 22].

Conjecture 7.6 (Holonomy-context principle). *For measure-dependent architectures such as Transformers, there exists a nonzero but finite optimal context-holonomy regime*

$$0 < \mathfrak{H}_{\text{opt}}^{\text{ctx}} < \infty$$

such that

- when $\mathfrak{H}^{\text{ctx}} \approx 0$, the network behaves as a nearly commutative context-averaging system;*
- when $\mathfrak{H}^{\text{ctx}}$ is too large, representation alignment deteriorates and trainability worsens;*
- the strongest in-context generalization occurs at an intermediate, nontrivial holonomy scale.*

Interpretation. Context dependence is conjectured to be a geometric consequence of changing transport rules, not merely of aggregating more tokens. The interacting-particle and geometric-dynamical perspectives on Transformers strongly motivate this view [4, 7].

Conjecture 7.7 (Spectral-gap collapse law). *For each attention layer, define the attention spectral gap*

$$\Delta_l = \sigma_1(A_l^{\text{attn}}) - \sigma_2(A_l^{\text{attn}}), \quad \Delta_L^{\text{int}} = \sum_{l=0}^{L-1} \Delta_l. \quad (160)$$

After controlling width, initialization, and operator norms, the integrated spectral gap Δ_L^{int} governs:

- (i) the rate of width-rank collapse,*
- (ii) the severity of depth-rank collapse,*
- (iii) the degradation of gradient propagation and the loss of task-bundle information.*

Interpretation. The attention spectral gap is conjectured to be a primary geometric control variable for rank-collapse phenomena in Transformers, building directly on recent spectral analyses of softmax attention [15].

Conjecture 7.8 (Hamiltonian core with dissipative shell). *A successful deep network contains an intermediate depth interval $[l_1, l_2]$ such that*

$$\delta_l^{\text{symp}} \text{ is small,} \quad \|S_l\|_{\text{op}} \text{ is small,} \quad \|\Omega_l\|_{\text{op}} \text{ is not small,} \quad l \in [l_1, l_2],$$

while shallower and deeper layers are more strongly dissipative. In this picture:

- *early layers perform coarse encoding and denoising;*
- *middle layers perform approximately symplectic or near-reversible long-range transport;*
- *late layers perform task-directed contraction, collapse, and readout.*

Interpretation. The conjecture is that useful deep architectures are neither globally symplectic nor globally dissipative. Instead, they possess a transport core whose geometry is closer to Hamiltonian propagation, surrounded by dissipative input and output shells [6, 10].

8 Technical Proof Sketches for Supporting Results

The hard theorem package in Section 6 is where the manuscript currently makes its strongest explanatory claim. The present section plays a different role. It records the mathematical backbone of the supporting results that surround that hard core.

What follows are proof sketches for Propositions 7.1–7.4. The goal is not to present final polished proofs, but to show that the extended Jacobian-chain program rests on coherent and standard mathematical arguments rather than on slogans alone.

8.1 Proof sketch for Supporting Proposition 7.1

Write the layer map in restricted form on E_l as

$$J_l|_{E_l} = (I + h_l A_l)|_{E_l}.$$

For any operator B with $\|B\|_{\text{op}} < 1$, the elementary singular-value inequalities

$$\sigma_{\max}(I + B) \leq 1 + \|B\|_{\text{op}}, \quad \sigma_{\min}(I + B) \geq 1 - \|B\|_{\text{op}}$$

hold. Applying these to $B = h_l A_l|_{E_l}$ yields

$$1 - h_l \|A_l|_{E_l}\|_{\text{op}} \leq \sigma_{\min}(J_l|_{E_l}) \leq \sigma_{\max}(J_l|_{E_l}) \leq 1 + h_l \|A_l|_{E_l}\|_{\text{op}}.$$

Because singular values of products satisfy

$$\sigma_{\max}(AB) \leq \sigma_{\max}(A)\sigma_{\max}(B), \quad \sigma_{\min}(AB) \geq \sigma_{\min}(A)\sigma_{\min}(B)$$

whenever the product is taken along compatible subspaces, iterating over l gives (152).

For the continuous-depth statement, let $J(t)$ solve

$$\dot{J}(t) = A(t)J(t), \quad J(0) = I,$$

and decompose $A(t) = S_t + \Omega_t$ with $S_t = S_t^\top$ and $\Omega_t^\top = -\Omega_t$. For a vector $v \in E_0$,

$$\frac{d}{dt} \|J(t)v\|^2 = 2 \langle J(t)v, A(t)J(t)v \rangle = 2 \langle J(t)v, S_t J(t)v \rangle,$$

since the skew part contributes zero to the quadratic form. Therefore

$$2\lambda_{\min}(S_t|_{E_t}) \|J(t)v\|^2 \leq \frac{d}{dt} \|J(t)v\|^2 \leq 2\lambda_{\max}(S_t|_{E_t}) \|J(t)v\|^2.$$

Grönwall’s inequality yields exponential upper and lower bounds for $\|J(t)v\|$, and taking the supremum over unit vectors in E_0 gives (153).

Why this matters. The proposition identifies the first-order part of the Jacobian-chain geometry that must be controlled in any architecture. In particular, it suggests that trainability diagnostics should be measured on task-restricted bundles, not only on the full ambient state space.

8.2 Proof sketch for Supporting Proposition 7.2

The proof is an iterated application of the second-order chain rule. For two C^2 maps F and G ,

$$D^2(F \circ G)(x)[u, v] = DF(G(x)) D^2G(x)[u, v] + D^2F(G(x))[DG(x)u, DG(x)v]. \quad (161)$$

Apply this identity first to $\Phi_{L-1} \circ \Phi_{L-2,0}$, then recursively expand the second term. Each recursion step inserts one new second-order contribution and sandwiches it between the Jacobians that occur before and after that layer. The result is exactly the sum in (154).

The norm bound (155) then follows by taking operator norms termwise:

$$\left\| \mathcal{J}_{L:l+1}(x_l) D^2\Phi_l(x_l) [\mathcal{J}_{l:0}u, \mathcal{J}_{l:0}v] \right\| \leq \|\mathcal{J}_{L:l+1}(x_l)\|_{\text{op}} \chi_l(x_l) \|\mathcal{J}_{l:0}u\| \|\mathcal{J}_{l:0}v\|.$$

Summing over l gives the stated inequality, and restricting u, v to the task bundle yields (156).

Why this matters. The formula makes it mathematically meaningful to say that a network becomes “too nonlinear.” The right culprit is not merely a large local Hessian, but a large local Hessian multiplied by the Jacobian chain that transports it.

8.3 Proof sketch for Supporting Proposition 7.3

The key classical fact is that a proper injective immersion is an embedding. Therefore, if $\Phi_l|_{M_l^c}$ is a proper injective immersion, then it identifies M_l^c diffeomorphically with its image $\Phi_l(M_l^c) = M_{l+1}^c$. In particular, all topological invariants preserved by diffeomorphism—connectedness, Betti numbers, homotopy type up to smooth equivalence—remain unchanged.

To obtain the proposition as stated, simply argue by contraposition. If any topological invariant changes between M_l^c and M_{l+1}^c , then $\Phi_l|_{M_l^c}$ cannot have been an embedding. There are only three generic ways this can fail in the present setting:

- (1) the differential loses rank, so immersion fails;
- (2) injectivity fails, so multiple points are identified;
- (3) the architecture explicitly imposes a quotient-like operation (pooling, masking, collapse, clipping, or another many-to-one map).

Why this matters. This proposition anchors topological language in concrete singular events of the Jacobian chain. It clarifies that one should not discuss holes, genus, or persistent homology in deep representations without simultaneously asking where rank loss or noninjectivity entered the network.

8.4 Proof sketch for Supporting Proposition 7.4

Choose a local coordinate chart on the quotient manifold $\tilde{\Theta} = \Theta/G$ near the class of θ_* , together with a smooth horizontal lift. In these coordinates, the projected update (157) becomes an update on a Euclidean chart with a position-dependent metric matrix G_θ and a Hessian compressed to the horizontal space.

Write $\theta_t = \theta_* + \delta_t$. Since θ_* is a critical point,

$$\nabla L(\theta_t) = H_{\theta_*}^{\text{hor}} \delta_t + O(\|\delta_t\|^2), \quad G_{\theta_t}^{-1} = G_{\theta_*}^{-1} + O(\|\delta_t\|).$$

Substituting these expansions into (157) gives

$$\delta_{t+1} = \delta_t - \eta G_{\theta_*}^{-1} H_{\theta_*}^{\text{hor}} \delta_t + O(\|\delta_t\|^2),$$

which is (158).

If $H_{\theta_*}^{\text{hor}}$ is self-adjoint in the G_{θ_*} -inner product, then the matrix

$$G_{\theta_*}^{-1/2} H_{\theta_*}^{\text{hor}} G_{\theta_*}^{-1/2}$$

is symmetric in ordinary Euclidean coordinates, hence has real eigenvalues. The linearized update is stable provided the spectral radius of

$$I - \eta G_{\theta_*}^{-1} H_{\theta_*}^{\text{hor}}$$

is strictly less than one, which is ensured by

$$0 < \eta < \frac{2}{\lambda_{\max}(G_{\theta_*}^{-1/2} H_{\theta_*}^{\text{hor}} G_{\theta_*}^{-1/2})}.$$

Why this matters. The proposition makes explicit that stable learning rates depend on geometry. Once parameter symmetries and statistical structure are accounted for, the relevant curvature quantity is horizontal and metric-dependent, not simply the largest eigenvalue of the raw Euclidean Hessian.

9 Experimental Program: Testing the Theory Rather Than Illustrating It

The experimental program is designed to test the theorem chain of the paper, not merely to decorate it with diagnostics. The central question is whether the geometric mechanism proposed here is genuinely predictive:

Do classwise compression, semantic-secant preservation, late-phase isotropy selection, and terminal alignment appear together for the reasons stated by the theory, and do they fail when the relevant geometric ingredients are selectively destroyed?

Accordingly, every empirical package in this section is organized around three principles.

- (P1) Interventions before correlations.** We do not regard post hoc correlation plots as decisive evidence. Every central quantity must be paired with a perturbation that is predicted to preserve it, weaken it, or destroy it.
- (P2) Theorem-level testing.** The relevant unit of validation is not an isolated observable but a theorem package: hypotheses, mechanism variables, and conclusion variables should move together.
- (P3) Breadth across model classes.** The collapse and optimization claims should be tested on controlled manifolds, residual vision models, and compact Transformer models, so that the theory is not mistaken for an artifact of one architecture family.

The current hard core of the paper yields four experimentally distinct claims.

- (C1) Collapse mechanism.** Theorem 6.6 and Corollaries 6.8–6.11 predict that classwise collapse is driven by nuisance-fiber contraction, semantic-secant preservation, and a controlled curvature floor.
- (C2) Terminal rigidity.** Theorem 6.12 and Corollary 6.13 predict that once the surviving semantic covariance becomes isotropic, the class means rigidify into simplex / ETF geometry and the classifier can lock onto the same structure.
- (C3) Late-phase selection.** Theorem 6.14, Theorem 6.16, and Corollaries 6.15–6.18 predict that balanced late training selects semantic isotropy rather than an arbitrary low-rank prototype cloud.
- (C4) Reduction from full training.** Theorem 6.21, Theorem 6.23, Theorem 6.26, and Corollaries 6.22, 6.25, and 6.27 predict that the reduced semantic mechanism is visible inside full-network late-phase training and small-step SGD.

The subsections below translate these claims into concrete experimental packages.

9.1 Package I: Collapse and prototype formation

This package is the empirical backbone of the manuscript. It tests whether collapse occurs precisely in the regime predicted by the bridge theorem.

Primary testbeds.

- **Controlled class-fiber data:** synthetic manifold-supported classification problems in which each class support has an interpretable within-class fiber structure.
- **Residual image classifiers:** ResNet-32/56/110, pre-activation ResNets, and moderate-scale vision encoders on CIFAR-10, CIFAR-100, Tiny-ImageNet, or ImageNet-100.

- **Compact Transformer classifiers:** small vision transformers or encoder-style Transformers on moderate-scale supervised classification tasks.

Primary interventions.

- (I1) strengthen or weaken nuisance-fiber contraction using Jacobian penalties on augmentation-derived within-class directions,
- (I2) preserve or disrupt semantic secants using class-center repulsion, class-mixing perturbations, or controlled semantic-secant regularizers,
- (I3) reduce or amplify the curvature floor through residual scaling, curvature penalties, normalization changes, or architecture modifications that alter χ_l .

Predicted signatures. Theorem 6.6 predicts that collapse should appear precisely when three geometric conditions occur together:

$$\begin{aligned} \rho_l &< 1 \text{ on nuisance fibers,} \\ \varepsilon_l^{\text{sem}} &\text{ small on semantic secants,} \\ \eta_l &\text{ small or summable across depth.} \end{aligned}$$

When these ingredients hold, the theory predicts:

- decay of the collapse ratio CR_l ,
- prototype concentration within each class,
- emergence of a nearest-class-center regime,
- explicit prototype-classifier error reduction.

By contrast, the theorem package predicts *structured failure modes*: strong contraction without semantic preservation should produce degenerate collapse; semantic preservation without contraction should preserve classification but leave a positive within-class floor; large curvature injection should destabilize or bend the representation before clean collapse forms.

9.2 Package II: Late-phase isotropy and terminal rigidity

This package asks what happens *after* prototype formation has begun. The issue is no longer whether classes collapse, but whether the surviving semantic skeleton becomes isotropic and then rigidifies into simplex / ETF geometry.

Primary testbeds. Balanced or nearly balanced classification runs are the preferred setting, since Theorem 6.14 and Theorem 6.16 are inherently class-symmetric statements. The most informative cases are runs that already display classwise concentration and therefore have clearly formed prototypes.

Primary interventions.

- (I1) preserve or break class balance through resampling, label imbalance, or class-dependent loss weights,
- (I2) preserve or break isotropization through prototype-level regularizers that either equalize or bias the semantic spectrum,
- (I3) preserve or break self-dual alignment through head regularization, head freezing, or controlled misalignment perturbations.

Predicted signatures. The terminal-rigidity and late-phase-selection theorems predict the following sequence:

$$\begin{aligned} \text{prototype concentration} &\implies \text{semantic isotropy} \\ &\implies \text{simplex / ETF rigidity} \\ &\implies \text{self-dual terminal alignment.} \end{aligned}$$

Empirically, the expected progression is:

- the semantic-isotropy defect δ_l^{iso} and the spectral Lyapunov surrogate $\mathcal{F}_l^{\text{iso}}$ decay in the late phase,
- the ETF deviation Δ_l^{ETF} tends to zero,
- the normalized reduced prototype margin $\hat{\mathbf{m}}_l$ saturates at the simplex value,
- the classifier-feature alignment defect decays once isotropy is present.

A key falsification test here is to preserve collapse while selectively breaking class balance or semantic isotropy. If ETF-type terminal geometry still appears generically under those perturbations, then the current late-phase selection mechanism is too narrow.

9.3 Package III: From full training to reduced semantic dynamics

This package addresses the main reduction claim of the paper: that the late semantic mechanism is not merely a reduced toy model, but is inherited from full-network cross-entropy training.

Primary testbeds. Runs should be monitored deep into the late phase on balanced or nearly balanced tasks for which class means, head weights, and class-averaged feature gradients can be estimated stably. Both gradient-flow proxies and small-step SGD runs are useful here.

Primary interventions.

- (I1) perturb semantic NTK balance or semantic Jacobian-frame homogeneity,
- (I2) perturb prototype concentration by weakening classwise compression,
- (I3) perturb head-span capture and self-dual alignment through head architecture choices, weight decay, or orthogonal head injections,
- (I4) compare full-batch, mini-batch, and small-step SGD regimes.

Predicted signatures. Theorems 6.21, 6.23, and 6.26 predict that, in the late phase,

- class-averaged feature gradients should become well approximated by reduced prototype gradients,
- the semantic NTK block should become close to a scalar identity on the semantic span,
- the empirical semantic Gram drift should track a reduced semantic flow,
- semantic anisotropy should dissipate directly along the full training trajectory,
- primitive head dynamics should eventually lock the classifier onto the semantic simplex once isotropy has formed.

This package is the strongest point of contact between the abstract geometric theory and ordinary deep-learning training dynamics.

9.4 Package IV: Extended diagnostics beyond the hard core

The remaining propositions and conjectures broaden the program. They are important, but they are not the main evidential bottleneck of the current manuscript.

Optimization geometry. Use deep linear or residual linear models to compare Euclidean, Fisher, and horizontal sharpness against maximal stable learning rates. This package tests Proposition 7.4 in the cleanest setting and asks whether quotient-information geometry improves on naive Euclidean curvature.

Curvature and topology. Use low-dimensional residual flows and synthetic manifold-supported data to compare curvature injection, restricted singular collapse, and persistent topology. This package tests Propositions 7.2 and 7.3.

Transformer-specific structure. Use synthetic sequence tasks and small language-model or encoder runs to test Conjectures 7.6 and 7.7, measuring whether context-sensitive performance depends on nontrivial holonomy and whether spectral-gap interventions control rank collapse. These tasks can also be used to test the broader hypothesis that middle layers of successful long-context models behave more like near-reversible transport blocks than late readout layers.

Recommended execution order. From a practical standpoint, the paper should be tested in the following order:

- (S1) controlled collapse experiments on synthetic class-fiber data,
- (S2) standard image-classification runs validating the bridge theorem and its immediate corollaries,
- (S3) balanced late-phase runs testing isotropy selection and terminal rigidity,
- (S4) full-to-reduced late-phase tracking on balanced image or language models,

(S5) only then the broader topology, optimization, and Transformer conjecture packages.

This order matches the present logical structure of the paper: first establish the hard core, then extend outward.

10 Observables and Estimation

The observables are organized by theorem package rather than by architecture family. The guiding principle is that every major formal statement in the paper should have a small set of primary readouts, together with a practical estimation recipe.

10.1 Package I observables: collapse, concentration, and simplified classification

The primary observables attached to Theorem 6.6 and Corollaries 6.8–6.11 are

$$\rho_l, \quad \varepsilon_l^{\text{sem}}, \quad \eta_l, \quad \text{CR}_l, \quad R_l, \quad \text{Tr}(W_l), \quad \text{Tr}(B_l), \quad d_l, \quad \widehat{\text{Err}}_l^{\text{NCC}}.$$

Here ρ_l is the nuisance-fiber contraction coefficient, $\varepsilon_l^{\text{sem}}$ is the semantic-secant distortion, η_l is the one-step curvature floor, $\text{CR}_l = \text{Tr}(W_l)/\lambda_{\min}^+(B_l)$ is the collapse ratio, R_l is the class-support radius, d_l is the minimum class-center separation, and $\widehat{\text{Err}}_l^{\text{NCC}}$ is the empirical nearest-class-center error.

Estimation.

- Estimate ρ_l from Jacobian action on augmentation-derived within-class directions or local tangent estimates inside class supports.
- Estimate semantic distortion by

$$\varepsilon_l^{\text{sem}} := \sup_{c \neq c'} \left| \frac{\|\mu_{l+1}^c - \mu_{l+1}^{c'}\|}{\|\mu_l^c - \mu_l^{c'}\|} - 1 \right|. \quad (162)$$

- Estimate the curvature floor via

$$\eta_l \asymp J_l \chi_l r_l^3 + \chi_l^2 r_l^4, \quad (163)$$

where J_l , χ_l , and r_l are estimated from Jacobian norms, second-order finite differences, and classwise fourth-moment radii.

- Estimate the collapse ratio by

$$\text{CR}_l = \frac{\text{Tr}(W_l)}{\lambda_{\min}^+(B_l)}, \quad (164)$$

computing $\lambda_{\min}^+(B_l)$ on the empirical semantic span or replacing it with a robust nonzero spectral floor when necessary.

- Estimate R_l by a classwise support radius or a robust upper quantile of $\|z - \mu_l^c\|$, and estimate d_l from pairwise class-center distances.

Why these are primary. These variables are the direct mechanism variables of the bridge theorem. If the hard core of the paper is correct, they should explain when collapse occurs and when classification simplifies.

10.2 Package II observables: isotropy, rigidity, and terminal geometry

The primary observables attached to Theorem 6.12, Theorem 6.14, Theorem 6.16, and Corollaries 6.15–6.18 are

$$\delta_l^{\text{iso}}, \quad \Delta_l^{\text{ETF}}, \quad \mathcal{F}_l^{\text{iso}}, \quad \widehat{\mathbf{m}}_l, \quad \mathcal{L}_{\alpha,l}^{\text{proto}}, \quad a_l^{\text{clf}}.$$

The semantic-isotropy defect is

$$\delta_l^{\text{iso}} := \frac{\|B_l - \beta_l P_l^{\text{sem}}\|_{\text{op}}}{\beta_l}, \quad \beta_l := \frac{\text{Tr}(B_l)}{K-1},$$

and the ETF deviation is

$$\Delta_l^{\text{ETF}} := \left\| \frac{1}{K\beta_l} M_l^\top M_l - \Pi_K \right\|_F.$$

We also use the equivalent spectral Lyapunov surrogate

$$\mathcal{F}_l^{\text{iso}} := \frac{1}{2} \left(\Delta_l^{\text{ETF}} \right)^2,$$

the normalized reduced prototype margin

$$\widehat{\mathbf{m}}_l := \min_c \min_{j \neq c} ((\widehat{G}_l)_{cc} - (\widehat{G}_l)_{jc}), \quad \widehat{G}_l := \frac{G_l}{\text{Tr}(G_l)},$$

and the fitted reduced prototype cross-entropy surrogate

$$\mathcal{L}_{\alpha,l}^{\text{proto}} := \frac{1}{K} \sum_{c=1}^K \log \left(1 + \sum_{j \neq c} \exp(-\alpha_l((\widehat{G}_l)_{cc} - (\widehat{G}_l)_{jc})) \right).$$

Finally, a_l^{clf} measures classifier–prototype alignment.

Why these are primary. These variables distinguish three stages that are often conflated in the literature: collapse, isotropization, and terminal alignment. The theory predicts that they occur in that order and for different geometric reasons.

10.3 Package III observables: reduction from full training to semantic dynamics

The primary observables attached to Theorems 6.21, 6.23, and 6.26 and Corollaries 6.22, 6.25, and 6.27 are

$$\Delta_t^{\text{sNTK}}, \quad \mathcal{E}_t^{\text{proto}}, \quad \mathcal{E}_t^{\text{flow}}, \quad \mathbf{c}_t^{\text{iso}}, \quad \mathcal{R}_t^{\text{iso}}, \quad \Lambda_t^\perp, \quad \mathcal{A}_t^{\text{head}}, \quad \nu_t.$$

Here Δ_t^{sNTK} is the semantic-NTK isotropy defect, $\mathcal{E}_t^{\text{proto}}$ is the feature-to-prototype reduction error, $\mathcal{E}_t^{\text{flow}}$ is the full-to-reduced flow-tracking residual, $\mathbf{c}_t^{\text{iso}}$ is the empirical anisotropy-coercivity ratio, $\mathcal{R}_t^{\text{iso}}$ is the anisotropy-dissipation residual, $\Lambda_t^\perp$ is the head-span leakage, $\mathcal{A}_t^{\text{head}}$ is the primitive self-dual alignment defect, and ν_t is the prototype drift rate.

Estimation.

- Estimate the semantic-NTK defect by

$$\Delta_t^{\text{sNTK}} := \frac{1}{\widehat{\kappa}_t} \left\| \widehat{\mathbb{K}}_t \Big|_{\widehat{E}_t^K} - \widehat{\kappa}_t I_{\widehat{E}_t^K} \right\|_{\text{op}}. \quad (165)$$

- Estimate the prototype-reduction error by

$$\mathcal{E}_t^{\text{proto}} := \max_c \left\| \widehat{g}_t^c - \nabla_{m^c} \mathcal{L}_{\widehat{\alpha}_t}(\widehat{G}_t) \right\|. \quad (166)$$

- Estimate the full-to-reduced flow-tracking residual by

$$\mathcal{E}_t^{\text{flow}} := \left\| \frac{d}{ds} \widehat{G}_t + \text{grad}_{\mathcal{G}_1} \mathcal{L}_{\widehat{\alpha}_t}(\widehat{G}_t) \right\|_F. \quad (167)$$

- Estimate the anisotropy-coercivity ratio and dissipation residual by

$$\mathbf{c}_t^{\text{iso}} := \frac{\left\langle \text{grad}_{\mathcal{G}_1} \mathcal{E}_{\text{iso}}(\widehat{G}_t), \text{grad}_{\mathcal{G}_1} \mathcal{L}_{\widehat{\alpha}_t}(\widehat{G}_t) \right\rangle_F}{\mathcal{E}_{\text{iso}}(\widehat{G}_t) + 10^{-12}}, \quad (168)$$

$$\mathcal{R}_t^{\text{iso}} := \frac{d}{ds} \mathcal{E}_{\text{iso}}(\widehat{G}_t) + \frac{\widehat{\lambda}_*}{2} \mathcal{E}_{\text{iso}}(\widehat{G}_t). \quad (169)$$

- Estimate head-span leakage, self-dual alignment, and prototype drift by

$$\Lambda_t^\perp := \|(I - P_t)H_t\|_F, \quad (170)$$

$$\mathcal{A}_t^{\text{head}} := \inf_{\alpha \geq 0} \|H_t - \alpha M_t\|_F, \quad (171)$$

$$\nu_t := \|\dot{M}_t\|_F. \quad (172)$$

Why these are primary. These quantities are what make the late-phase story falsifiable. Without them, “reduction to semantic dynamics” would remain a narrative instead of a measurable claim.

10.4 Supporting observables: transport, curvature, topology, and Transformer structure

The remaining observables support the broader Jacobian-chain program.

Restricted transport. Measure

$$\alpha_l, \quad \beta_l, \quad \kappa_l^{\text{task}}, \quad \kappa_l^{\text{nuis}},$$

using JVP/VJP routines on estimated task and nuisance subspaces, with nuisance directions extracted from label-preserving augmentations and task directions from class-mean differences and local logit gradients.

Local strain and twisting. Measure

$$s_l^+ := \lambda_{\max}(S_l), \quad s_l^- := \lambda_{\min}(S_l), \quad \omega_l := \|\Omega_l\|_F.$$

These diagnose local amplification, contraction, and non-gradient-like twisting.

Curvature injection. Measure

$$\chi_l, \quad \mathfrak{C}_l,$$

with second-order finite-difference approximations such as

$$D^2\Phi_l(z)[u, u] \approx \frac{J_l(z + \varepsilon u)u - J_l(z)u}{\varepsilon}. \quad (173)$$

Holonomy and noncommutativity. Measure

$$\mathfrak{H}_l = \sum_{i < j \leq l} \| [A_i, A_j] \|_F^2, \quad \mathfrak{H}_l^{\text{ctx}} = \left\| P_l (J_l^{(c_1)} J_l^{(c_2)} - J_l^{(c_2)} J_l^{(c_1)}) P_{l-1} \right\|_F^2.$$

Optimization geometry. Measure the sharpness triplet

$$\lambda_{\max}^{\text{Euc}}(H), \quad \lambda_{\max}^{\text{Fisher}}(F), \quad \lambda_{\max}^{\text{hor}}.$$

Transformer-specific spectral observables. Measure

$$\Delta_l = \sigma_1(A_l^{\text{attn}}) - \sigma_2(A_l^{\text{attn}}), \quad \text{srnk}(A_l^{\text{attn}}), \quad \text{srnk}(X_l), \quad \lambda_{\text{ang}}, \quad \lambda_{\text{grad}}.$$

Topology and symplectic structure. Measure

$$\beta_k^\varepsilon(M_l^c), \quad \text{Dgm}_k(M_l^c), \quad \tau_l^{\text{topo}}, \quad \sigma_l^{\text{sing}}, \quad \delta_l^{\text{symp}}.$$

These observables support Propositions 7.2–7.3 and Conjecture 7.8.

11 What Would Count as Success?

The paper should be evaluated theorem-package by theorem-package. Strong evidence means that the full implication chain of a package appears under the interventions predicted by the theory and disappears under targeted counter-interventions. Weak evidence means that some conclusion variable behaves as predicted without a clean demonstration of mechanism. Clear falsification means that the advertised conclusion appears robustly even when the package’s geometric premises are selectively broken.

11.1 Package I: bridge theorem and immediate deep-learning consequences

What would support it. Strong support for Theorem 6.6 and Corollaries 6.8–6.11 would look like this:

- runs with $\rho_l < 1$, small $\varepsilon_l^{\text{sem}}$, and controlled η_l are precisely the runs in which CR_l decays,
- the same runs show prototype concentration, decreasing within-class covariance, and improving nearest-class-center performance,
- perturbing any one of contraction, semantic preservation, or curvature control selectively destroys the predicted consequence.

What would count against it. The bridge theorem would be in trouble if collapse regularly appears without semantic-secant preservation, if semantic preservation plus weak contraction still yields strong within-class collapse, or if the prototype-classifier error bound fails systematically in the regime where the theorem’s hypotheses appear numerically satisfied.

11.2 Package II: terminal rigidity and late-phase selection

What would support it. Strong support for Theorem 6.12, Theorem 6.14, Theorem 6.16, and Corollaries 6.15–6.18 would require:

- semantic-isotropy defect δ_l^{iso} decays before or together with ETF deviation Δ_l^{ETF} ,
- the normalized prototype margin $\widehat{m}_l$ approaches the simplex value in exactly the runs with vanishing ETF deviation,
- the reduced prototype cross-entropy surrogate behaves as a monotone selector for the same endpoint,
- class-balance or symmetry-breaking interventions destroy isotropy selection and prevent ETF-type terminal geometry.

What would count against it. This package would be weakened if ETF geometry appears generically without an isotropization stage, if the margin does not distinguish ETF-forming runs from non-ETF runs, or if severe class imbalance leaves the same terminal geometry essentially unchanged.

11.3 Package III: reduction from full training to reduced semantic dynamics

What would support it. Strong support for Theorems 6.21, 6.23, and 6.26 and Corollaries 6.22, 6.25, and 6.27 would require:

- small prototype-reduction error $\mathcal{E}_t^{\text{proto}}$,
- small semantic-NTK defect Δ_t^{sNTK} ,
- small full-to-reduced flow residual $\mathcal{E}_t^{\text{flow}}$,
- positive anisotropy-coercivity ratio $\mathbf{c}_t^{\text{iso}}$ together with small dissipation residual $\mathcal{R}_t^{\text{iso}}$,
- decay of head-span leakage $\Lambda_t^\perp$ and self-dual defect $\mathcal{A}_t^{\text{head}}$ once isotropy has formed and prototype drift ν_t is small,
- small-step SGD inheriting the same terminal selector as the gradient-flow proxy.

What would count against it. This reduction package would be seriously weakened if late-phase full-network training reaches the advertised terminal geometry while the reduced semantic dynamics fails to track the trajectory, if SGD systematically selects a different semantic endpoint than the gradient-flow proxy in the same regime, or if head alignment appears without any decay of span leakage or isotropy-dependent locking.

11.4 Package IV: supporting propositions and broader conjectures

Optimization geometry. Proposition 7.4 is supported if Fisher or horizontal sharpness predicts maximal stable learning rates more accurately than Euclidean sharpness. It is weakened if Euclidean sharpness consistently dominates once symmetries and output statistics are controlled.

Curvature and topology. Propositions 7.2 and 7.3 are supported if curvature accumulation predicts instability better than raw Jacobian norm alone and if topological transitions cluster near restricted singular collapse or explicit quotient operations. They are weakened if topology changes appear with no nearby Jacobian singularity signature.

Selective isometry. Proposition 7.1 and Conjecture 7.5 are supported if restricted singular values explain trainability, collapse, and robustness better than full-space Jacobian diagnostics, and if the best-collapse runs are precisely those with near-isometric semantic directions and strongly contractive nuisance directions.

Transformer structure. Conjectures 7.6 and 7.7 are supported if context-sensitive performance is maximized in an intermediate nonzero holonomy regime and if spectral-gap interventions jointly reduce width-rank collapse, depth-rank collapse, and gradient pathology. They are weakened if these phenomena decouple systematically.

Hamiltonian core. Conjecture 7.8 is supported if middle layers are measurably more symplectic than early or late layers and if that signature correlates with stronger long-range information transport. It is weakened if the symplectic defect carries no such structure.

A final standard. The paper should be judged successful only if the theorem packages survive *counterfactual* tests. In other words, a good run is not enough; what matters is whether the proposed geometric mechanism is the thing that makes the good run good.

12 Figures and Tables for the Final Paper

The visual material in the final paper should do more than decorate the text. It should help the reader see the logic of the manuscript, understand which theorem package is being tested, and identify which geometric quantities matter for deep learning in practice. The figures and tables should therefore mirror the architecture of the paper rather than simply list diagnostics one by one.

Figure group I: stating the problem. These figures should tell the reader, in visual form, what the paper is trying to explain.

- **Figure 1: A unified geometric picture of deep learning.** A single schematic showing the four spaces used throughout the paper—representation manifold, context / measure space, parameter quotient space, and statistical manifold—together with the deep-learning phenomena associated with each one.
- **Figure 2: The hard theorem chain.** A compact visual of the central implication

$$\begin{aligned} \text{local Jacobian geometry} &\implies \text{classwise compression} \implies \text{intermediate collapse} \\ &\implies \text{semantic isotropy} \implies \text{terminal simplex / ETF rigidity} \implies \text{self-dual alignment.} \end{aligned}$$

This should function as the one figure that immediately tells a reader what the mathematical spine of the paper is.

Figure group II: testing the hard core. These figures should directly correspond to the main theorem package.

- **Figure 3: Bridge-theorem dashboard.** Layerwise or epochwise trajectories of ρ_l , $\varepsilon_l^{\text{sem}}$, η_l , and CR_l on the same run. The point is to show that nuisance-fiber contraction, semantic-secant preservation, curvature control, and collapse-ratio decay actually move together.
- **Figure 4: Collapse and prototype formation.** Within-class covariance, between-class covariance fraction, class radius, prototype concentration, and nearest-class-center error shown together. This figure should make the passage from geometric conditions to prototype-like features visually unmistakable.
- **Figure 5: Terminal rigidity and isotropy selection.** The semantic isotropy defect, ETF deviation, normalized prototype margin, and classifier–feature alignment plotted on a shared horizontal axis. This is the cleanest visual test of the late-phase route to terminal neural collapse.

Figure group III: optimization and architecture. These figures should show that the framework is not merely descriptive but predictive for training and intervention.

- **Figure 6: Sharpness notions against stable training.** Euclidean sharpness, Fisher sharpness, and quotient-aware sharpness plotted against maximal stable learning rate or edge-of-instability diagnostics.
- **Figure 7: Transformer phase diagram.** Horizontal axis Δ_L^{int} , vertical axis $\mathfrak{H}^{\text{ctx}}$, color indicating trainability, collapse severity, or task accuracy. This figure belongs to the extended Jacobian-chain program beyond the hard core.
- **Figure 8: Full-flow to reduced-flow tracking.** Late-phase trajectories of Δ_t^{sNTK} , $\mathcal{E}_t^{\text{proto}}$, $\mathcal{E}_t^{\text{flow}}$, and the semantic-isotropy defect on a common run. The purpose is to show that the reduced selector is visible in full training rather than being only a formal reduction.

Tables. The tables should function as navigation devices for the theory.

- **Table 1: Theorem-to-observable map.** Each theorem, corollary, proposition, or conjectural package paired with the main observables and the experiment intended to test it.
- **Table 2: Intervention-to-effect map.** Fiber-contraction regularization, semantic-secant preservation, spectral-gap projection, curvature regularization, symplectic regularization, gauge fixing, and geometry-aware optimization, together with the predicted geometric effect and the theorem package each intervention is meant to probe.
- **Table 3: Assumption ledger.** A compact table separating what is proved, what is reduced-model, what is late-phase, and what remains conjectural. This would substantially improve readability for mathematically sophisticated readers.

13 Discussion, Limitations, and Open Problems

The manuscript is now organized around a definite hard core rather than a flat list of ideas. The main mathematical contribution is a bridge from local Jacobian-chain geometry to global class statistics and then to neural-collapse-type terminal structure. This matters because it connects differential geometric quantities that can be measured locally—restricted Jacobians, curvature floor, anisotropy defects, symmetry-corrected sharpness—to large-scale representation phenomena that matter directly in deep learning. At the same time, several parts of the present theory remain reduced, late-phase, or partially conjectural, and it is important to say clearly where the manuscript is strongest and where it is not yet final.

What is established. At its strongest, the paper now explains one concrete chain of deep-learning phenomena. It shows why successful features become prototype-like without losing class separation, and how a further late-phase isotropization of the surviving semantic covariance can force the representation toward regular-simplex / ETF geometry. This is more than a geometric reinterpretation of known neural-collapse observations. The hard theorem package identifies a mechanism: within-class contraction, semantic-secant preservation, and curvature control produce prototype concentration; isotropy of the residual semantic covariance produces rigidity; and reduced or approximate late-phase optimization selects that isotropy. In this sense, the paper contributes a mechanism theorem rather than only a descriptive framework.

What the theory gives deep learning. For deep learning, the value of the theory is threefold. First, it explains *why* good representations simplify: they contract nuisance fibers while preserving semantic secants. Second, it turns that explanation into measurable diagnostics, including collapse ratio, semantic-secant distortion, curvature floor, anisotropy defect, normalized prototype margin, and quotient-aware sharpness. Third, it points to actionable interventions: architecture, normalization, regularization, and optimization choices can be evaluated by whether they move the system toward the geometric regime predicted by the hard theorem chain. This is the practical payoff of the framework.

Where the current manuscript is strongest. The paper is currently most mature on the representation side. The bridge theorem, the prototype-concentration and nearest-class-center

corollaries, the rigidity upgrade to simplex / ETF geometry, and the late-phase selector all belong to one coherent narrative. The theorem package is now strong enough that the rest of the paper can be read as support, extension, or challenge to that central chain. This is a qualitatively stronger position than a draft that merely proposes terminology or diagnostics.

Where the assumptions are still strong. Several assumptions remain stronger than one would want in a final definitive theory. First, the task / nuisance decomposition is not intrinsic in a representation-independent sense; it must be estimated from supervision, data geometry, or augmentation structure. Second, the late-phase reduction results still rely on conditions such as prototype concentration, semantic-frame homogeneity, contractive reduced-head dynamics, or semantic NTK rigidity. These assumptions are motivated and partially bootstrapped, but not yet derived from raw SGD in a fully end-to-end manner. Third, the Transformer, holonomy, and symplectic parts of the program are less complete than the collapse side. They belong naturally to the Jacobian-chain framework, but they are not yet part of the hardest proved spine of the article.

What remains reduced or conjectural. The reduced semantic-Gram dynamics, max-min-margin selection principles, and primitive head flows should be understood as mathematically clean late-phase models. They are valuable because they isolate the correct variables and make sharp statements possible. But they remain reductions, and the long-term task is to derive them from original-network dynamics under weaker and more realistic assumptions. Likewise, the holonomy, spectral-gap, topology, and Hamiltonian-core directions are now well-posed and measurable, but they remain outside the central proved backbone of the present manuscript.

Main open problems. The next questions are now sharply defined.

- Can semantic-secant near-isometry be derived directly from architectural or optimization assumptions rather than estimated after training?
- Can the coercive anisotropy-dissipation conditions be deduced from full cross-entropy / Fisher geometry in a realistic late-phase regime?
- Can semantic NTK rigidity be replaced by more primitive Jacobian or feature-homogeneity assumptions in the full-to-reduced theorem?
- Can the task / nuisance splitting be made more intrinsic, perhaps through label-conditioned Fisher geometry or semantic kernel geometry?
- Can the context-holonomy and spectral-gap mechanisms be brought to the same theorem level as the collapse side of the paper?
- Can these ideas be turned into architecture-selection or training-control rules that improve stability and generalization at scale?

Overall assessment of the current state. The manuscript should now be read as a theory paper with a genuine core theorem package and a wider geometric program built around it. It is no longer only proposing vocabulary. It already explains one important chain of deep-learning phenomena. At the same time, it is not yet the final word: the late-phase reductions still need to be weakened, the optimization bridge should become more primitive, and the extended Jacobian-chain

directions should gradually move from conjectural diagnostics to proved mechanisms. That is exactly the right place for the paper to be at this stage.

14 Conclusion

This paper develops a geometric foundation for one concrete deep-learning question: how a trainable network turns raw data into classwise denoised, semantically separated, easy-to-classify features without losing optimization stability. The answer proposed here is that the relevant mathematical object is the Jacobian chain. Its restricted first-order action governs semantic transport and nuisance contraction, its second-order action governs nonlinear distortion, and the quotient-information geometry induced by the model governs meaningful sharpness and stable optimization scales.

The strongest contribution of the current manuscript is now a connected theorem package rather than a collection of isolated observations. The hard core shows that within-class fiber contraction, semantic-secant preservation, and a summable curvature floor imply prototype formation and intermediate collapse; that late-phase isotropization of the residual semantic covariance forces the surviving class means toward regular-simplex / ETF geometry; and that reduced late-phase optimization, together with increasingly primitive head and flow reductions, provides a principled route by which training can select that terminal geometry. In short,

$$\begin{aligned} \text{local Jacobian geometry} &\implies \text{classwise compression} \implies \text{intermediate collapse} \\ &\implies \text{semantic isotropy} \implies \text{simplex / ETF rigidity and self-duality.} \end{aligned}$$

For deep learning, this has a direct meaning. The theory explains why successful features become prototype-like, why simple prototype classifiers become effective on late representations, which geometric quantities should be measured to distinguish benign collapse from instability, and which architectural or optimization interventions should matter if one wants to stabilize transport, simplify representations, or control late-phase geometry. This is the practical benefit of the framework: it turns representation, collapse, and optimization into parts of one mathematical picture that can be tested, falsified, and used.

The manuscript is therefore best viewed as a current, substantially hardened version of a broader program. It already contains a real mathematical backbone; it also leaves behind a precise frontier. The next frontier is to push more of the late-phase assumptions back into primitive end-to-end training dynamics and to bring the extended Jacobian-chain directions—holonomy, spectral collapse, topology, and symplectic transport—up to the same theorem level as the collapse core. That is the path from the present manuscript to a final, submission-ready theory of geometric structure in deep learning.

A Appendix Roadmap

A natural appendix structure for a full paper version is:

- **Appendix A.** Detailed proofs of Propositions [7.1–7.4](#).
- **Appendix B.** Practical estimation of task and nuisance bundles.
- **Appendix C.** Efficient computation of JVPs, VJP, HVPs, and restricted singular values.

- **Appendix D.** Persistent-homology subsampling, computational complexity, and robustness checks.
- **Appendix E.** Construction of context holonomy observables for Transformer experiments.
- **Appendix F.** Quotient geometry, gauge fixing, and symmetry-aware sharpness computations.

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